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Dynamic Volunteer Staffing in Multicrop Gleaning Operations

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Abstract. Gleaning programs organize volunteer gleaners to harvest a variety of leftover crops that are donated by farmers for the purpose of feeding food-insecure individuals. Thus, the gleaning process simultaneously reduces food waste and food insecurity. However, the operationalization of this process is challenging because gleaning relies on two uncertain sources of input: the food and labor supplies. The purpose of this paper is to help gleaning organizations increase the (value-weighted) volume of fresh food gleaned by better managing the uncertainties in the gleaning operation. We develop a model to capture the uncertainties in food and labor supplies and seek a dynamic volunteer-staffing policy that maximizes the payoff associated with the amount of food gleaned. The exact analysis of the staffing problem seems intractable. Therefore, we resort to an approximation in the heavy traffic regime. In that regime, we characterize the system dynamics of the multicrop gleaning operation and derive the optimal staffing policy in closed form. The optimal policy is a nested threshold policy that specifies the staffing level for each class of donation (i.e., a donation of a particular crop type and donation size). The policy depends on the number of available gleaners and the backlog of gleaning donations. A numerical study using data calibrated from a gleaning organization in the Boston area shows that the dynamic staffing policy we propose can recover approximately 10% of the volume lost when the gleaning organization uses a static policy. To achieve this improvement, no capital or major process changes would be required—only some small changes to the staffing level requests.

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Keywords: gleaning • volunteer • food bank • food waste • food insecurity • dynamic control

1. Introduction

The practice of gleaning combines two societal problems to create an elegant solution for both. Gleaning dates back to ancient times when landowners allowed the poor to gather crop left on the fields after the harvest. This act of generosity alleviated the dual paradoxical problems of food waste and food insecurity. These problems persist and are arguably even magnified in modern times. The amount of food wasted in the United States is staggering—almost 40% of food fit for human consumption is wasted (Gunders 2012). A significant portion of this waste occurs at the farm level when farmers leave crops unharvested because market conditions make it unprofitable to pay for harvesting labor. For example, the crop could be blemished or misshapen but perfectly edible and nutritious, or a bumper crop season could drive the market price down. A USDA estimate indicates that 6% of fruit and vegetable crops are left unharvested (USDA 2011).

Although edible and healthy food was left unharvested in farm fields, 14% of U.S. households in 2014 were food insecure. The USDA definition of food insecurity is "a household-level economic and social condition of limited or uncertain access to adequate food" (USDA 2015). This translates to 17.4 million households in which 15.3 million members are children (Coleman-Jensen et al. 2015). It is not simply calories lacking in food-insecure households, but insufficient consumption of micronutrient-rich foods, such as fruits and vegetables (Davis and Tarasuk 1994, Tarasuk and Beaton 1999). A number of studies have found links between food insecurity and health-related issues, such as obesity, diabetes, and hypertension (Gomez et al. 2013, Weinfield et al. 2014). Thus, it is not just increasing the quantity of food but increasing the quantity of the right type of food that will help alleviate the problems associated with food insecurity. In particular, healthy foods, such as fresh fruits and

vegetables, can help improve and balance the diets of food-insecure individuals.

Juxtaposing these two societal problems, it seems a natural solution would be to use the unharvested crops that would otherwise go to waste to help food-assistance programs (e.g., food banks) provide healthy, nutritious food to supplement the diet of food-insecure individuals. In fact, many gleaning organizations nationwide have already implemented this solution. For example, in recent years, gleaning programs in California, Ohio, Arizona, Texas, and New York have distributed 127, 26, 18.5, 13, and 3.6 million pounds, respectively, of fruits and vegetables on an annual basis (California Association of Food Banks 2011, Schuelke et al. 2011, Vitiello et al. 2015).

Although the gleaning solution to food waste and food insecurity is conceptually elegant and appealing, the operationalization of the solution poses distinct challenges. Gleaning relies on two uncertain sources of input: the food and labor supplies. Gleaning donations arrive randomly as farmers themselves face uncertainty in crop yield, market prices, and labor supply. Donations can be of varying crop types (e.g., apples, lettuce) and of various sizes (measured in pounds of produce). Although the overall gleaning season can be six months to year-round, depending on geographic location, when the gleaning organization is notified of a gleaning donation (i.e., communication from a farmer for an opportunity to glean an unharvested crop), it must schedule a gleaning trip within a short window of time, usually on the order of a few days to a week, to reap the crop before it perishes on the field. Moreover, to make gleaning economically feasible, gleaning organizations use volunteer gleaners to harvest the crop. Using historical gleaning trip data and current communication with its volunteer pool, the gleaning organization typically has a good idea of how many gleaners are available at a given time. However, because these gleaners are volunteers, they are not obliged to attend a scheduled gleaning trip even if they are available and sign up for the trip. Therefore, the labor supply is uncertain.

Clearly, gleaning organizations exist and are successful in their mission to provide healthy food to food-assistance programs. There are many ways to measure the performance of a gleaning operation. One metric is the total volume, that is, the total number of pounds gleaned. However, because of the inherent differences in crop types, there may be differences in "value" per pound across crop types. For example, value could be measured in servings, in which case carrots would have 2.5 servings per pound, whereas lettuce would have 6.3 servings per pound. Value could also be measured in the market value of the crop. The purpose of this paper is to help these organizations increase the value of fresh food gleaned by better managing the uncertainties in a gleaning operation.

We develop a model to capture the uncertainties in food and labor supply and derive a dynamic volunteer-staffing policy that maximizes the long-run average value of food gleaned. Our policy gives gleaning organizations a simple staffing policy to manage a key trade-off that arises because of the gleaning operation's reliance on volunteer labor. On the one hand, when a gleaning donation arrives, the organization would like to maximize the volume gleaned (which translates to a particular value) on the scheduled trip. Each gleaning trip is typically three hours long. Each gleaner is able to harvest a certain volume of a given crop type; therefore, the more volunteer gleaners that attend a trip, the higher the output volume (up to the total volume available for harvesting). This would imply requesting a high volunteer-staffing level—even higher than the number of gleaners needed to harvest the available crop because some volunteers may not show up. Note, however, that the expected marginal output from each unit increase in volunteer-staffing level is decreasing. Moreover, unlike employees who can be required to attend every gleaning trip, volunteers who do show up will likely not attend another gleaning trip for some time. This may cause another gleaning opportunity that arises in the near future to have low volunteer gleaner attendance and low output volume or even require the gleaning organization to turn down the donation completely because low volunteer availability would delay the scheduling of the trip too far into the future. This could lead to loss of farmer goodwill. Therefore, on the other hand, the organization may want to request a lower volunteer-staffing level and sacrifice output on the current trip in order to "save" volunteer labor for anticipated future gleaning trips. The trade-off between immediate and future output is crucial, especially when gleaner utilization is high, that is, when the gleaning donation rate is high relative to the rate at which volunteers become available.

In fact, the staffing-level trade-off faced by the gleaning organization is similar to the trade-off in the model, that is, the cost versus the benefit of a marginal unit of capacity. However, there is a crucial difference. In the newsvendor model, the shadow price for the marginal unit of capacity is determined by exogenous parameters, such as demand characteristics (or arrival rate of customers), cost, and penalty values. In our setting, the shadow price of a marginal unit of capacity (i.e., an increase in staffing level) also depends on the staffing levels set in previous periods. That is, the shadow price arises endogenously based on the staffing policy deployed. Because of the distinct nature of this process, a dynamic model is required to capture the staffing-decision trade-off.

We characterize the system dynamics of the gleaning operation and derive the optimal staffing policy that maximizes the gleaning organization's net payoff, that is, the total value of gleaned produce minus penalties incurred for turning down donations (e.g., loss of farmer goodwill). We show that the optimal policy is a nested threshold policy that specifies the staffing level for each class of donation (i.e., a donation of a particular crop type and donation size). The policy depends on the number of available gleaners and the backlog of gleaning donations. Our characterization of the thresholds is almost explicit in the sense that the calculations involved are lightning fast; they only involve finding the roots of strictly monotone functions. We calibrate our model based on volunteer and gleaning trip data from the Boston Area Gleaners (BAG) (2016). Using the calibrated model, we perform a simulation study that shows the dynamic policy improves the net payoff compared with two policies used in practice: the static policy and the unlimited policy. The magnitude of the improvement depends on the penalty for rejecting donations. Within a reasonable range of penalty values, our results showed a net payoff improvement range from approximately 3% to 6% compared with the static policy and on the order 40% compared with the unlimited policy. This modest improvement in net payoff translates to the recovery of approximately 10% of the volume lost using the static policy and approximately 75% of the volume lost using the unlimited policy. To put this in perspective, for a gleaning organization around the size of the Boston Area Gleaners (2016) using the static policy, the recovered volume would be equivalent to approximately 18,500 (four-ounce) servings per year.

2. Literature Review

Topically, this paper is related to a stream of literature that studies operations that support humanitarian relief organizations. Much of the research in this area focuses on "sudden onset" disasters, such as earthquakes and hurricanes (see Altay and Green 2006, Kovacs and Spens 2011, Galindo and Batta 2013, and Starr and Van Wassenhove 2014 for reviews). However, there is emerging work on "slow onset" disasters that are not triggered by a specific event, but are severely adverse conditions that create significant hardship for vulnerable segments of the population (Van Wassenhove 2006). The systemic problem of food insecurity is an example of such slow onset disasters. There is a stream of extant literature that focuses on nonprofit food bank operations (see Johnson and Smilowitz 2007 for an overview). Most of the research in this area studies tactical decisions in food bank operations, such as vehicle routing for food collection and distribution (Gunes et al. 2010, Yildiz et al. 2013, Davis et al. 2014, Lien et al. 2014, Solak et al. 2014), warehouse operations (Mohan et al. 2013), and forecasting techniques for food donation (Davis et al. 2013). In the literature that focuses on food bank gleaning operations, much of the work studies the impact of gleaning programs on recipients' diet in terms of nutrition and preference (e.g., Teron and Tarasuk 1999, Irwin et al. 2007) and improving food security (Hoisington et al. 2001). The papers most related to ours are Sönmez et al. (2016) and Lee et al. (2017) that examine the operational efficiency of a food bank gleaning operation. Sönmez et al. (2016) and Lee et al. (2017) are focused on a particular field study and use a simulation-optimization approach to compare the effectiveness of a given feasible set of fixed trip schedules. In contrast, we model a general setting and derive an (almost) explicit characterization of an optimal dynamic staffing policy.

Similar to our paper, the literature in humanitarian operations has also studied volunteer management. These papers focus on specific settings in which volunteer task and scheduling criteria have been critical (see Sampson 2006 for an overview). For example, Falasca and Zobel (2012) developed a spreadsheet model for optimizing volunteer schedules that consider volunteer preferences and task/schedule desirability. Similarly, Gordon and Erkut (2004) developed a scheduling program for a music festival staffed by volunteers. In our setting, the organization cannot schedule volunteers to a particular shift. The challenge is to manage the uncertainty in labor capacity. We develop a model and derive an optimal policy that is generalizable to other volunteer settings in which the attendance of volunteers is uncertain.

The gleaning process studied in this paper has similar characteristics to operations in the sharing economy and inventory settings. The research on ride-sharing operations studies queues with self-scheduling capacity. This setting is similar to ours in that the number of servers is stochastic. However, a key difference from our context is that the servers in the sharing economy are compensated, and thus, a key managerial lever for managing capacity is setting the compensation level. Banerjee et al. (2015) compared the performance of dynamic and static pricing schemes when there is imperfect information on system parameters. Taylor (2018) investigated how delay sensitivity and agent independence affects the optimal price and server compensation. Ibrahim (2018) jointly optimized the staffing, compensation, and delay announcement decision of a system with self-scheduling servers. Dong and Ibrahim (2017) provide recommendations on when a firm should use a blended workforce (consisting of full-time and (a random number of) contingent workers that are paid different wage levels).

An inventory setting that is related to our paper is the inventory control problem with random yields;

Yano and Lee (1995) provide a comprehensive description of the issues in this context and a review of the literature. The similarity of the random-yield setting and ours is the problem of how to set capacity or inventory quantity, that is, balancing the overage and underage costs when the availability of the relevant resource (inventory or volunteers) is uncertain. Using a modified newsvendor model with demand uncertainty and production yield uncertainty, early papers focused on the single-period setting, characterizing the optimal order policy, which includes whether to place an order depending on the level of on-hand inventory and, if so, what the production/order quantity should be (cf. Karlin 1958, Giffler 1960, Shih 1980, Gerchak et al. 1988, Henig and Gerchak 1990).

A more complex problem addressed in the random-yield inventory is the multiperiod problem in which inventory can be carried over from period to period. Gerchak et al. (1988) show that the solution to the multiperiod random-yield problem is not myopic, and Henig and Gerchak (1990) proved the structure of the optimal policy in a general setting (including showing that the order point is higher when the yield is random than when it is certain). Deriving the optimal policy is difficult, and the solution approach has been to develop heuristics to determine the optimal order quantity. (cf. Mazzola et al. 1987, Bollapragada and Morton 1999, Zipkin 2000, Inderfurth and Transchel 2007, Huh and Nagarajan 2010, Inderfurth and Vogelgesang 2013, Inderfurth and Kiesmuller 2015). Unlike the inventory setting in which the current order quantity is not upper bounded by previous orders, in the volunteer-staffing problem, the staffing level requested in the past affects the availability of volunteers currently available. Moreover, unlike the inventory setting in which demand and decisions are periodic, donation arrival timing is random, and volunteer availability must be synchronized with the arrivals to capture the value of the fresh produce as in a service setting. Thus, we use a queueing model to capture the dynamics of the volunteer gleaning process, and the structure of the optimal policy prescribes positive or negative changes to staffing level to adjust for system congestion. In a similar inventory setting, Huang et al. (2008) studied the optimal policy when there is replacement warranty demand, which depends on demand from previous periods and reduces the capacity to serve the new demand in a given period. A key difference in Huang et al. (2008) is that the reduction in capacity is a fixed fraction of previous demand, whereas volunteer availability in our setting is random.

The methodology used in this paper contributes to a stream of literature that studies dynamic resource allocation under uncertainty. At a high level, the abstract mathematical model developed in this paper is related to the literature on (dynamic) service rate control in Markovian queueing systems. Many papers have been written on different versions of the service-rate control problem over the past 50 years (cf. Crabill 1972, 1974; Weber and Stidham 1987; Stidham and Weber 1989; George and Harrison 2001; see Stidham 1988, 2002 for a survey through circa 2002). More recently, motivated by wireless communications applications, Ata (2005) and Ata and Zachariadis (2007) studied this question for a queueing system with a finite buffer in which congestion concerns are captured by imposing a constraint on the fraction of blocked jobs. Ata (2005) focused on choosing the service rate dynamically as a function of the queue length. The cost rate associated with the chosen service rate is convex increasing. Ata and Zachariadis (2007) extended this to a setting in which the environment is changing randomly. Both papers derive closed-form formulas for the optimal (dynamic) service rate as a function of the queue length. Ata and Shneorson (2006) considered a setting in which both the arrival and service rates are chosen dynamically and characterize the optimal policy in closed form. Hopp et al. (2007) also examined a setting in which the service rate can be chosen dynamically and compared it with the fixed service-rate case.

Our problem has several important differences from those studied by the aforementioned papers. Arguably, the most important difference is that the service rate (or staffing level for each gleaning donation) chosen in the current period affects the service rate in future periods directly because the volunteers working now will not be available for some time. This process feature is central to the planner's optimization problem.

From a methodological perspective, our work is also related to the control of queueing systems in heavy traffic, pioneered by Harrison (1988). In this literature, one approximates the queueing system with a Brownian model, which is simpler to analyze (see, for example, Harrison and Wein 1989, 1990; and Wein 1991). The validity of this approach and the accuracy of the resulting approximation are verified in the literature by simulation studies and heavy traffic-limit theorems (see, e.g., Harrison 1998, Bell and Williams 2001, and Ata and Kumar 2005). Several papers in the heavy traffic literature formulate control problems with delay constraints and derive more tractable approximations by replacing the delay constraints with upper bounds on the queue lengths in the heavy traffic regime as done in this paper (see, e.g., Plambeck et al. 2001, Ata 2006, and Rubino and Ata 2009). We follow the same approach and approximate a challenging control problem with a simpler one in the heavy traffic limit. However, the limiting regime we use is the one advanced in the seminal paper, Halfin and Whitt (1981), in which the arrival rate and the number of servers grows simultaneously; also see Browne and Whitt (1995) for a study of related diffusion processes.

The asymptotic analysis of our problem gives rise to a drift-rate control problem. There have been several papers in recent years that consider controlling the drift rate of a diffusion. Ata et al. (2005) studied the drift-rate control of a reflected diffusion on a bounded interval under general costs of the drift rate chosen; also see Ata (2006), Ghosh and Weerasinghe (2010), Ormeci-Matoglu and Vande Vate (2011), Ata and Tongarlak (2013), Kim and Ward (2013). In two closely related papers, Rubino and Ata (2009) and Ghamami and Ward (2013) considered scheduling parallel-server queueing systems. Rubino and Ata (2009) considered optimal scheduling and admission control for a general parallel-server system with abandonments under linear holding costs, abandonment penalties, costs for rejecting jobs, and delay constraints, which are replaced with bounds on queue lengths in the asymptotic formulation. Rubino and Ata (2009) solved the drift-rate control problem arising in the heavy traffic regime, proposed a policy for the general parallel-server system, and showed its effectiveness through simulations. Ghamami and Ward (2013) considered a similar setting and also provided a proof of the asymptotic optimality of their proposed policy.

Among the drift-rate control problems mentioned, the closest to ours is Ata et al. (2005). To be specific, we can use the approach in Ata et al. (2005) whenever there is a positive backlog of gleaning donations and all volunteers are busy. However, when there are available volunteers (i.e., the negative part of the state space), their approach breaks down because the system dynamics are very different in the positive part of the state space. Thus, a novel method is needed to analyze the problem on the negative part of the state space, which we develop. Moreover, combining the solutions on the negative and positive parts of the state space in a smooth fashion is far from straightforward and requires us to do the "smooth pasting" with ad hoc arguments from first principles. Hence, our paper also contributes to the literature of diffusion-control problems by explicitly solving the volunteer-staffing problem.

3. The Gleaning Operation and the Planner's Dynamic Staffing Problem

We represent the gleaning operation as a queueing process. The process starts with a donation from a farmer. Each donation is of a particular crop type, indexed by $i = 1, \ldots, I$ (e.g., apples, lettuce, etc.), and a particular size s_{ij}, $j = 1, \ldots, J_i$, measured in pounds of produce. The combination of crop type and the size of donation will define a class. Thus, there are $J_1 + J_2 + \ldots + J_I$ classes of donations. We use the double subscript ij to denote the class of a donation. Gleaning donations from farmers of class ij arrive randomly according to a Poisson process of rate λ_{ij} to the gleaning organization; we denote the arrival process of class ij gleaning donations by $\{R_{ij}(t); t \geq 0\}$. We refer to the gleaning organization as the *planner*. When the gleaning donation arrives, the planner schedules a gleaning trip that is staffed by volunteer gleaners who harvest the crop. We refer to volunteer gleaners as *volunteers*.

The entire volunteer pool consists of n volunteers. Each volunteer is capable of harvesting α_i pounds of produce i on a typical three-hour gleaning trip. Each pound of produce has value v_i, which can be measured in dollars, number of servings, or another appropriate measure of value. Thus, on a three-hour trip, each volunteer's gleaning output generates a payoff of $r_i = v_i\alpha_i$. To harvest the entire crop associated with a gleaning donation of class ij requires $\tilde{k}_{ij} = s_{ij}/\alpha_i$ volunteers. Therefore, each gleaning donation has a potential total payoff of $r_i\tilde{k}_{ij}$, but the realized value is the minimum of the potential payoff and r_i multiplied by the number of volunteers that attend the gleaning trip.

Volunteers, unlike employees, are not obliged to attend every scheduled gleaning trip. Typically, multiple gleaning trips are scheduled per week during the gleaning season (gleaning seasons are usually six to 12 months long). A few very active volunteers may attend a trip every week, whereas many volunteers only attend once or twice a season. Therefore, after an active harvesting time of three hours, a volunteer typically has a period of repose that may last weeks to several months during which the volunteer is not available for gleaning trips. We consider the combination of the harvest and repose periods as a volunteer's total service time—during this service time, the volunteer is *unavailable* for signing up for gleaning trips. Note that the harvest time for each volunteer is identical, but the repose times differ; moreover, the harvesting time of three hours is negligible compared with the repose period. At the end of the entire service time (harvest plus repose), the volunteer becomes *available* to attend gleaning trips. Because the gleaning organization typically has a good idea of how many gleaners are available (using historical data and through communication with volunteers), we assume that the number of available volunteers is observable. The average rate that a volunteer becomes available after repose is denoted by μ, and we model the service time of a volunteer as an exponential random variable.

When a gleaning donation arrives, the planner requests a volunteer-staffing level, k. The request for k volunteers can be sent to the volunteer pool via an online sign-up portal similar to the one used by the Boston Area Gleaners (2016). Typically, getting volunteers to sign up if they are available is fairly easy. Once the staffing-level request is made, those volunteers who are available sign up for the gleaning trip slots, up to the requested staffing level. Thus, we assume that if a staffing level of k is requested, k volunteers will sign up. However, because volunteers

are not employees, there is some chance that they will not actually show up even if they signed up for the trip. We use q to denote the probability that a volunteer attends a gleaning trip once signed up and $N_{ij}(k)$ to denote the number of volunteers that attend a class ij gleaning trip when the requested staffing level is k. Assuming volunteers act independently, we model $N_{ij}(k)$ as a binomial random variable with k trials and a success probability of q. Thus, for a class ij gleaning trip with requested staffing level k, the expected payoff is

$$\Pi_{ij}(k) = r_i \mathbb{E}[\min\{N_{ij}(k), \tilde{k}_{ij}\}]. \tag{1}$$

Because a class ij gleaning trip uses $N_{ij}(k)$ volunteer servers, we make a flow unit conversion from gleaning donation (trip) to volunteer jobs by breaking down each class ij gleaning donation into $N_{ij}(k)$ volunteer *jobs*.

The service time for a job is the harvest time plus repose time, that is, the entire time the volunteer is unavailable. If there are not enough volunteers available when a gleaning donation arrives, our model backlogs the jobs, not necessarily the entire donation, unless there are no available volunteers. Thus, this backlog is expressed in units of volunteer jobs; that is, one backlogged gleaning donation of class ij is equivalent to a backlog of $N_{ij}(k)$ volunteer jobs if the staffing level k is requested. The planner decides the staffing level for gleaning donations of various classes dynamically over time. That is, the planner chooses a dynamic staffing policy $\{k_{ij}(t) : t \geq 0\}$ for all i, j. Given the planner's dynamic staffing policy $\{k_{ij}(t) : t \geq 0\}$ for all i, j, the cumulative number of volunteer jobs (to be served) arriving by time t, denoted by $A(t)$, is given as follows:

$$A(t) = \sum_{i=1}^{I} \sum_{j=1}^{J_i} \int_0^t N_{ij}(k_{ij}(s))\, dR_{ij}(s), \quad t \geq 0.$$

We use $Q(t)$ to denote the number of jobs currently in the gleaning operation at time t and call this the queue-length process. This includes the number of volunteers who are currently unavailable (i.e., actively harvesting or in repose) and the number of virtual jobs that are backlogged. It is easy to see that the process $\{Q(t) : t \geq 0\}$ corresponds to a Markovian queue with arrival process $\{A(t) : t \geq 0\}$ and n exponential servers. Thus, its dynamics can be described by its infinitesimal transition rates.

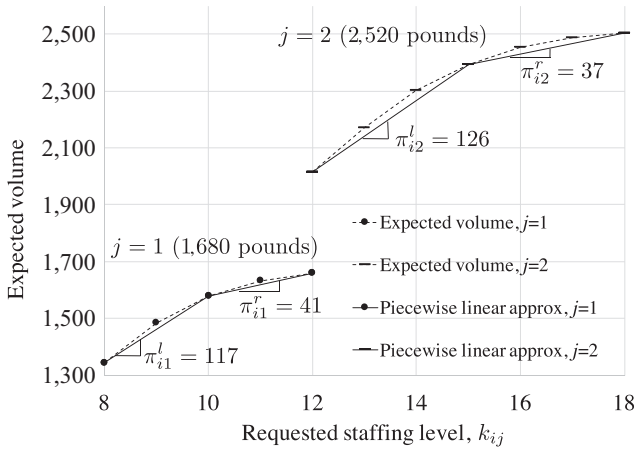
Gleaning is a time-sensitive process because crops must be harvested in a timely manner. When a farmer calls with a gleaning donation, the gleaning trip must be scheduled within a few days to at most a week or else the crop will perish on the field. In other words, there is a constraint on the delay experienced by the backlogged donations. To meet such delay constraints, the planner may choose to reject gleaning donations upon arrival. That is, if the number of backlogged jobs is too high (i.e., there are too many donations already in the queue), the lead time to the scheduled trip will be too long and the planner will have to reject the gleaning donation. If a donation is rejected, the planner not only loses the expected payoff of the trip, but also incurs a loss of farmer goodwill. Farmers are willing to work with gleaning organizations, but they expect a certain level of consistency from the organization. If the gleaning organization often rejects gleaning donations, then farmers will be less willing to expend the extra effort to engage with the organization in the future. To capture that the planner loses more than just the payoff of the current donation when rejecting it, we represent the loss of farmer goodwill by a penalty $\tilde{p}_{ij}$ (e.g., in potential future donations) when a class ij gleaning donation is rejected.

The expected payoff of the gleaning trip (Equation (1)) clearly increases the staffing level, k. There are two factors that drive this increase. First, there is a productivity effect: each volunteer is only capable of generating r_i in value. Therefore, the marginal attending volunteer adds r_i to the total payoff of the trip, up to $\tilde{k}_{ij}$ volunteers on a given trip. Second, there is a safety-capacity effect: with probability $1 - q$, a volunteer who signs up does not show up for the trip. Up to a nominal staffing level k^*_{ij}, the productivity effect dominates. Beyond k^*_{ij}, the safety-capacity effect dominates. Thus, the expected payoff function for a class ij donation increases steeply in the requested staffing level k_{ij} up to the nominal staffing level k^*_{ij}, then increases less steeply as the productivity effect diminishes and marginal volunteers are only useful if there are no-shows (i.e., the safety-capacity effect).

Figure 1 shows the expected payoff curves (dotted lines) as a function of the requested staffing level, k_{ij}, for a donation of crop type $i = 2$ (apples) with two donation sizes $j = 1$ (1,680 pounds) and $j = 2$ (2,520 pounds). In the figure, $q = 0.8$, $v_i = 1$, and $\alpha_i = 210$ pounds. For $j = 1$, if $\tilde{k}_{i1} = 8$ volunteers are required to harvest the entire crop of 1,680 pounds, then setting a nominal staffing level of $k^*_{i1} = \tilde{k}_{i1}/q$ would imply that in expectation, eight volunteers would show up. The staffing level range is set to ±20% of the nominal (i.e., $k^*_{i1} \pm 0.2k^*_{i1}$) so that $k_{i1} \in [\underline{k}_{i1}, \bar{k}_{i1}]$, where $\underline{k}_{i1} = 8$ and $\bar{k}_{i1} = 12$. Similarly, the staffing level range for $j = 2$ is set to $k^*_{i2} \pm 0.2k^*_{i2}$ such that $\underline{k}_{i2} = 12$ and $\bar{k}_{i2} = 18$.

The payoff functions in Figure 1 show the difference between the productivity effect and the safety-capacity effect of the marginal volunteer. For each expected payoff function, $j = 1, 2$, to the left of the nominal staffing level k^*_{ij}, the slope of the payoff function is steep (productivity effect), whereas to the right of k^*_{ij}, the slope flattens out (safety-capacity effect). We approximate the expected payoff function as a piecewise linear function (solid lines in Figure 1). We approximate the two pieces of the function using the slopes

Figure 1. Expected Payoff Functions and Piecewise Linear Approximations for Two Donation Classes of Crop Type $i = 2$ (Apples) with Two Donation Sizes $j = 1$ (1,680 Pounds) and $j = 2$ (2,520 Pounds); $q = 0.8$, $v_i = 1$, $\alpha_i = 210$ pounds; $\bar{k}_{i1} = 8$, $k^*_{i1} = 10$, $\underline{k}_{i1} = 8$, $\bar{k}_{i1} = 12$; $\bar{k}_{i2} = 12$, $k^*_{i2} = 15$, $\underline{k}_{i2} = 12$, $\bar{k}_{i2} = 18$



$\pi^l_{ij} = (\Pi_{ij}(k^*_{ij}) - \Pi_{ij}(\underline{k}_{ij}))/(k^*_{ij} - \underline{k}_{ij})$ and $\pi^r_{ij} = (\Pi_{ij}(\bar{k}_{ij}) - \Pi_{ij}(k^*_{ij}))/(\bar{k}_{ij} - k^*_{ij})$. We use such a piecewise linear approximation of the expected payoff to derive the optimal staffing policy. However, to assess the performance of the policy in the simulation study in Section 6, we use the realizations of the payoff of each trip.

The planner seeks to maximize the total value generated through gleaning. Because of volunteer attendance uncertainty, the planner may want to build some safety capacity in the requested staffing level (k_{ij} for class ij). The downside to this is that there may be excess volunteers on an early trip and not enough volunteers for a later trip. In fact, the planner may have to reject future gleaning donations if volunteer availability drops too low, resulting in long delays—this would further reduce the volume of crops gleaned. This trade-off is crucial when the system is critically loaded, that is, when it is in heavy traffic. In other words, the demand rate for volunteers is close to the rate at which they become available.

Given the trade-off between increasing the value of crops gleaned on the immediate trip versus a future trip, the planner can potentially improve the long-run average performance by dynamically adjusting the volunteer-staffing level based on the congestion in the system. If there is low congestion, then the planner can afford to request a high staffing level because the shadow price for volunteer capacity is low. If congestion is high, the planner needs to be more frugal with volunteer capacity and request a lower staffing level. To state the planner's problem, let π_{ij} denote the piecewise linear approximation of the expected payoff Π_{ij} on the relevant range of $[\underline{k}_{ij}, \bar{k}_{ij}]$ (as illustrated in Figure 1). That is, for $k_{ij} \in [\underline{k}_{ij}, \bar{k}_{ij}]$ and nominal staffing level k^*_{ij}, we define

$$\pi_{ij}(k) = \begin{cases} \Pi_{ij}(k^*_{ij}) + \pi^r_{ij}(k_{ij} - k^*_{ij}), & k_{ij} \in [k^*_{ij}, \bar{k}_{ij}], \\ \Pi_{ij}(k^*_{ij}) + \pi^l_{ij}(k_{ij} - k^*_{ij}), & k_{ij} \in [\underline{k}_{ij}, k^*_{ij}]. \end{cases} \tag{2}$$

Within a reasonable staffing level range $[\underline{k}_{ij}, \bar{k}_{ij}]$, the planner's problem is to find the optimal control $k_{ij}(t) \in [\underline{k}_{ij}, \bar{k}_{ij}]$ for $t \geq 0$ and all i, j, such that the staffing level maximizes the net payoff over the long run. To be more specific, for each class ij, the planner chooses the dynamic staffing policy $\{k_{ij}(t) \in [\underline{k}_{ij}, \bar{k}_{ij}] : t \geq 0\}$ so as to maximize the average net payoff (i.e., maximize the total payoff from scheduled trips, taking into account the effect of rejection penalties), subject to the delay constraints of the backlogged gleaning donations. Unfortunately, this formulation is not tractable analytically. Therefore, in the next section, by considering a sequence of closely related systems in heavy traffic, we formulate an approximate yet far more tractable formulation.

4. An Approximating Brownian Control Problem

The dynamic staffing problem is not tractable in its *exact* form as discussed earlier. Therefore, by considering a sequence of closely related systems indexed by the total number of volunteers n, we formulate a more tractable approximation. To be specific, we study this sequence of systems as the number of volunteers n gets large under the so-called heavy traffic assumption; a superscript n is attached to the quantities of interest in the nth system. In particular, λ^n_{ij} is the average arrival rate of class ij gleaning donations, and $\mu^n = n\mu$ is the average rate at which volunteers become available in the nth system.

The traffic intensity ρ^n of the nth system is given by $\rho^n = \sum_{i=1}^{I} \sum_{j=1}^{J_i} \lambda^n_{ij} k^*_{ij} q/(n\mu)$. Next, we state the assumption on how the traffic intensity ρ^n changes as the number of volunteers increases.

Assumption 1 (Heavy Traffic Assumption). *We assume that $\rho^n < 1$ for all n and that*

$$\lim_{n\to\infty} (1 - \rho^n)\sqrt{n} = \beta > 0. \tag{3}$$

Letting $\rho^n_{ij} = \lambda^n_{ij} k^*_{ij} q/(n\mu)$ for all ij, we also assume that $\lim_{n\to\infty} \rho^n_{ij} = \rho_{ij} > 0$, where ρ_{ij} is defined as the limit on the left-hand side. In particular, we have that

$$\sum_{i=1}^{I} \sum_{j=1}^{J_i} \rho_{ij} = 1. \tag{4}$$

We focus on this regime because, when the volunteer gleaner capacity is close to the demand rate (resulting from the arrival rate of gleaning donations), the planner benefits most from being able to dynamically adjust the

staffing policy. If the volunteer gleaning capacity was much higher than the demand rate, the planner could always request a high staffing level without worrying about adversely affecting the volunteer attendance on future gleaning trips. On the other hand, if volunteer gleaning capacity were much lower than the demand rate, then the planner would always set safety capacity to zero because there would always be a future gleaning trip when the volunteer capacity would be productively used. In the regime in which the relative values of volunteer capacity and the demand rate are close so that the demand rate can sometimes be greater than volunteer capacity because of fluctuations in volunteer availability, the ability to adjust the staffing level based on the number of jobs in the system can have a significant impact on the output.

As mentioned, the planner dynamically adjusts the requested staffing level, denoted by $k_{ij}(t)$ for each class ij of gleaning donations and $t \geq 0$. We focus our attention on the following functional form for $k_{ij}^n(t)$:

$$k_{ij}^n(t) = k_{ij}^* \left(1 - \frac{\theta_{ij}(t)}{\sqrt{n}}\right), \quad t \geq 0, \; i = 1, \ldots, I, \; j = 1, \ldots J_i, \quad (5)$$

where $\theta_{ij}(\cdot)$ is the control for class ij. This functional form is natural because the congestion is of second order relative to the system size as the number of volunteers n gets large in the heavy traffic regime. Thus, as the backlog in the system varies, only second order adjustments to the requested staffing level per trip are warranted. In particular, the adjustment proposed in Equation (5) to the nominal staffing level k_{ij}^* is $k_{ij}^* \theta_{ij}(t)/\sqrt{n}$ in the nth system, and the assumed decay rate of $1/\sqrt{n}$ translates to a drift-rate reduction in the approximating Brownian control problem.

Recall that $Q^n(t)$ denotes the number of jobs currently in the gleaning operation at time t, which includes the number of volunteers currently unavailable and the number of virtual jobs that are backlogged. We define the scaled version of $Q^n(t)$ as follows:

$$Z^n(t) = \frac{Q^n(t) - n}{\sqrt{n}}, \quad t \geq 0. \quad (6)$$

If $Z^n(t) > 0$, it means there are backlogged jobs and no available volunteers, whereas if $Z^n(t) < 0$, then there is no backlog of jobs and there are volunteers available. The infinitesimal drift and variance of $Z^n(\cdot)$, denoted by m_n and σ_n^2, respectively, are defined as follows:

$$m_n(z, t) = \lim_{h \downarrow 0} \mathbb{E}\left[\frac{Z^n(t+h) - Z^n(t)}{h} \mid Z^n(t) = z\right],$$
$$\sigma_n^2(z, t) = \lim_{h \downarrow 0} \mathbb{E}\left[\frac{(Z^n(t+h) - Z^n(t))^2}{h} \mid Z^n(t) = z\right],$$

see chapter 15 of Karlin and Taylor (1981).[1]

Defining $\sigma^2 = \mu\left[2 + q\sum_{i=1}^{I} \sum_{j=1}^{J_i} \rho_{ij}(k_{ij}^* - 1)\right]$ and

$$m(z) = \begin{cases} -\mu z, & z < 0, \\ 0, & z \geq 0, \end{cases}$$

the following proposition characterizes the asymptotic behavior of m_n, σ_n.

Proposition 1. *We have that* $\lim_{n\to\infty} \sigma_n^2(z, t) = \sigma^2$ *and*

$$\lim_{n\to\infty} m_n(z, t) = m(z) - \mu(\beta + \sum_{i=1}^{I} \sum_{j=1}^{J_i} \rho_{ij}\theta_{ij}(t)).$$

In light of Proposition 1, as done in theorem 2 of Halfin and Whitt (1981), one can check Stone's criterion (see for example, theorem 3.2 of Iglehart 1965) to conclude that Z^n converges weakly to a diffusion process with the drift rate and variance parameters as characterized in Proposition 1.

Because crops must be harvested in a timely manner, another important feature of the volunteer-staffing problem is that the planner has congestion concerns and wishes to limit the delay between the time a donation arrives and the time at which gleaning occurs. The delay constraint is not really meaningful in a conventional queueing model because the delays are random variables. Nonetheless, it becomes meaningful in an asymptotic sense. That is, it is met with probability one because the law of large numbers implies an equivalence between (scaled) bounds on the number of jobs in the system and (scaled) bounds on delays as one approaches the heavy traffic limit. Therefore, one tractable surrogate model for the delay constraint is to impose an upper bound on the queue-length process[2] as done in Plambeck et al. (2001), Ata (2005), and Rubino and Ata (2009). This corresponds to imposing an upper bound on the backlog of jobs; that is,

$$Q^n(t) - n \leq b^n. \quad (7)$$

In the asymptotic regime we consider, the deviations of the queue-length process Q^n from n are of order $\sqrt{n}$. So we assume that in the nth system, we have that

$$b^n = \sqrt{n}b,$$

where $b > 0$ is a parameter to be chosen by the planner. Note that by Little's law and the heavy traffic assumption, the upper bound imposed on the backlog corresponds to delays of order $b/(\mu\sqrt{n})$. This can be used to calibrate b so as to incorporate the planner's congestion concerns.

To impose an upper bound on the queue-length process, the planner turns away donations when the queue length reaches its upper bound, exercising admission

control. Turning away a donation of class ij at time t results in a rejection penalty of $\tilde{p}_{ij}$ and turning away $N_{ij}(k^n_{ij}(t))$ volunteer jobs, where

$$\mathbb{E}[N_{ij}(k^n_{ij}(t))] = qk^n_{ij}(t) = qk^*_{ij}\left(1 - \frac{\theta_{ij}(t)}{\sqrt{n}})\right) \approx qk^*_{ij}. \quad (8)$$

In what follows, we approximate the number of volunteer jobs turned away with each such class ij donation that is rejected by qk^*_{ij}, ignoring the lower order (random) terms. This is a reasonable approximation because our method of analysis is crude enough that such differences are washed away under the heavy traffic scaling as we pass to the limit.

Moreover, for mathematical convenience, we model the admission control as if the planner can turn off the clock for the arrival process (when the queue length reaches its upper bound). In particular, we let $U^n_{ij}(t)$ denote the cumulative amount of time that the arrival process is turned off for class ij during the interval $[0, t]$.

As mentioned, we consider a sequence of closely related systems indexed by the number of volunteers n whose formal limit is the Brownian control problem. The original dynamic staffing problem can be viewed as a specific element of this sequence of problems, which is determined by the number of volunteers. The underlying assumption of the Brownian approximation is that the number of volunteers (for the original problem of interest) is large enough so that various (scaled) performance-relevant processes of the original system can be approximated by the corresponding processes of the Brownian control problem.

Passing to the limit formally as the number of volunteers n gets large, we assume that[3]

$$\begin{aligned}&(Z^n(\cdot), \sqrt{n}\, U^n_{11}(\cdot), \ldots, \sqrt{n}\, U^n_{IJ_I}(\cdot))\\ &\quad \Rightarrow (Z(\cdot), U_{11}(\cdot), \ldots, U_{IJ_I}(\cdot)) \text{ as } n \to \infty, \end{aligned} \quad (9)$$

where $\Rightarrow$ denotes weak convergence; see Billingsley (1999).

Given the rejection and arrival processes of class ij denoted by $U^n_{ij}(\cdot)$ and $R^n_{ij}(\cdot)$, respectively, we approximate the cumulative number of donations of class ij rejected by time t, that is, $R^n_{ij}(t) - R^n_{ij}(t - U^n_{ij}(t))$, as follows:

$$R^n_{ij}(t) - R^n_{ij}(t - U^n_{ij}(t)) \approx \lambda^n_{ij} U^n_{ij}(t) \approx \frac{\sqrt{n}\mu\rho_{ij}}{k^*_{ij}q}\, U_{ij}(t),$$

where we used Equation (9) and the approximation $\lambda^n_{ij} \approx n\mu\rho_{ij}/(k^*_{ij}q)$; see Assumption 1 for a justification of the latter approximation. Then using the approximation in Equation (8), we approximate the cumulative number of volunteer jobs of class ij rejected until time t by $\sqrt{n}\mu\rho_{ij}\, U_{ij}(t)$. Scaling this further by $\sqrt{n}$, in accordance with the scaling of Z^n (see Equation (6)), and defining

$$U(t) = \sum_{i=1}^{I}\sum_{j=1}^{J_i} \rho_{ij} U_{ij}(t), \quad t \geq 0,$$

and building on Proposition 1, we assume that the processes Z and U jointly satisfy the following:

$$\begin{aligned} dZ(t) = &\left[m(Z(t)) - \mu\left(\beta + \sum_{i=1}^{I}\sum_{j=1}^{J_i}\rho_{ij}\theta_{ij}(t)\right)\right]dt \\ &+ \sigma\, dB(t) - \mu\, dU(t), \quad t \geq 0, \end{aligned} \quad (10)$$

$$Z(t) \leq b, \quad t \geq 0, \quad (11)$$

$$\int_0^\infty \mathbb{1}_{\{Z(t) < b\}}\, dU(t) = 0, \quad (12)$$

$$U(\cdot) \text{ is nondecreasing with } U(0) = 0, \quad (13)$$

where B is the standard Brownian motion.

To repeat, the controlled process Z described in Equation (10) is (the limit of) the scaled version of the queue-length process Q^n. Similarly, U_{ij} is (the limit of) the scaled version of the cumulative rejection process U^n_{ij}. Equation (13) states the natural constraint that U must be nondecreasing, whereas Equation (12) ensures that it increases only when the (scaled) backlog Z reaches its upper bound b.

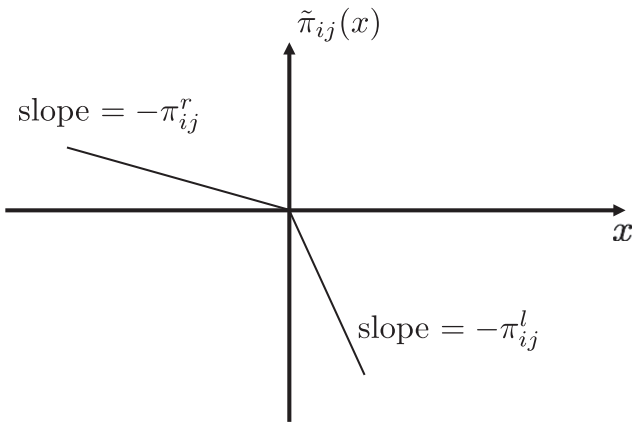
Equation (11) imposes an upper bound on the backlog of jobs Z, ensuring that delays do not become excessive. Note, however, that the backlog $Z(t)$ is not observable to the planner at time t because it is the sum of the number of volunteer jobs for all of the currently backlogged gleaning donations. Moreover, the number of volunteer jobs for each gleaning donation is not known until the trip for the donation takes place. Only then does the planner know if a volunteer shows up or not. However, $Z(t)$ can be approximated closely in the asymptotic regime by multiplying each donation by the expected number of volunteers attending given the requested staffing level for the donation and summing over all donations in the backlog.

Equations (10)–(13) describe the evolution of the system state in the approximating Brownian control problem given the controls $\theta_{ij}(\cdot)$ for all ij. To complete the description of the approximating problem, it only remains to specify its objective. To this end, recall our approximation $\pi_{ij}(k)$ from Equation (2). Then we let $\tilde{\pi}_{ij}(x) = \pi_{ij}(k^*_{ij} - x) - \pi_{ij}(k^*_{ij})$ for $x \in [k^*_{ij} - \bar{k}_{ij}, k^*_{ij} - \underline{k}_{ij}]$. That is,

$$\tilde{\pi}_{ij}(x) = \begin{cases} -\pi^r_{ij}x, & x \in [k^*_{ij} - \bar{k}_{ij}, 0], \\ -\pi^l_{ij}x, & x \in [0, k^*_{ij} - \underline{k}_{ij}]. \end{cases}$$

Figure 2 displays an illustrative $\tilde{\pi}_{ij}(\cdot)$ function.

Figure 2. An Illustrative $\tilde{\pi}_{ij}(\cdot)$ Function



To facilitate the analysis to follow, we make the following assumption; see Online Appendix C.2 for a discussion of how to relax this assumption.

Assumption 2. *We have that $\tilde{p}_{ij} = pk^*_{ij}$, where $p > 0$ is a parameter.*

Online Appendix C.3 provides an approximation of the planner's expected payoff. Building on that approximation and defining

$$\underline{\theta}_{ij} = \frac{\sqrt{n}(k^*_{ij} - \bar{k}_{ij})}{k^*_{ij}} < 0 \text{ and}$$
$$\bar{\theta}_{ij} = \frac{\sqrt{n}(k^*_{ij} - \underline{k}_{ij})}{k^*_{ij}} > 0 \text{ for all } i, j,$$

we arrive at the following formulation for the planner's problem: for $s \geq 0$, choose $\theta_{ij}(s) \in [\underline{\theta}_{ij}, \bar{\theta}_{ij}]$ for all i, j so as to

$$\text{maximize} \quad \underline{\lim}_{t\to\infty} \frac{1}{t}\, \mathbb{E}\left[\int_0^t \frac{\mu}{q} \sum_{i=1}^{I} \sum_{j=1}^{J_i} \tilde{\pi}_{ij}(\rho_{ij}\theta_{ij}(s))\, ds - p\frac{\mu}{q}\, U(t)\right]$$

subject to (10)–(13).

To facilitate the analysis to follow, define $\hat{\pi}_{ij}(x) = -\tilde{\pi}_{ij}(x)$ for $x \in [\underline{\theta}_{ij}, \bar{\theta}_{ij}]$ and all i, j. Then we formulate the following problem: choose $\theta_{ij}(s) \in [\underline{\theta}_{ij}, \bar{\theta}_{ij}]$ for all $s \geq 0$ and i, j so as to

$$\text{minimize} \quad \overline{\lim}_{t\to\infty} \mathbb{E} \left[\frac{1}{t}\int_0^t \sum_{i=1}^{I} \sum_{j=1}^{J_i} \mu\, \hat{\pi}_{ij}(\rho_{ij}\theta_{ij}(s))\, ds + \mu p\, U(t)\right] - \mu \sum_{i=1}^{I} \sum_{j=1}^{J_i} \pi^r_{ij}\rho_{ij}\underline{\theta}_{ij} \qquad (\tilde{P})$$

subject to (10)–(13).

It is straightforward to argue that this formulation is equivalent to the previous one. To be more specific, to arrive at ($\tilde{P}$), one multiplies the objective of the previous one by $-q$ and formulates the problem as one of minimization. Finally, the constant term $\sum_{i=1}^{I}\sum_{j=1}^{J_i} \pi^r_{ij}\rho_{ij}\underline{\theta}_{ij}$ is subtracted from the objective. None of these mathematical operations change the nature of the problem nor its optimal solution. However, this transformation helps us streamline the analysis to follow. Let $\mathscr{A} = \{(i, j, d) : i = 1, \ldots, I, j = 1, \ldots, J_i, d = l, r\}$.

Assumption 3. *The elements of the set $\{\pi^d_{ij} : (i, j, d) \in \mathscr{A}\}$ are all distinct.*

Remark 1. Otherwise, one may combine the identical values "suitably" to arrive at a case in which the assumption is satisfied. ■

Let $M = 2(J_1 + J_2 + \ldots + J_I)$ and consider the set $\{\pi^d_{ij} : (i, j, d) \in \mathscr{A}\}$ and let $\text{rank}(\pi^d_{ij})$ denote the rank of π^d_{ij} among the elements in that set. In addition, for $m = 1, \ldots, M$, let $i(m), j(m)$ denote the crop type and the size of the donation, respectively, for the mth smallest element, that is, the element with rank m. Similarly, let $d(m) \in \{l, r\}$ denote whether the mth smallest element corresponds[4] to the "left slope" or the "right slope."

In particular, let c_m denote the mth smallest element of the set $\{\pi^d_{ij} : (i, j, d) \in \mathscr{A}\}$. That is,

$$c_m = \pi^{d(m)}_{i(m), j(m)} \text{ for } m = 1, \ldots, M. \qquad (14)$$

Similarly, for $m = 1, \ldots, M$, let

$$\eta_m = \begin{cases} \rho_{i(m), j(m)}\bar{\theta}_{i(m), j(m)} & \text{if } d(m) = l, \\ \rho_{i(m), j(m)}|\underline{\theta}_{i(m), j(m)}| & \text{if } d(m) = r. \end{cases}$$

To facilitate the analysis to follow, define $\underline{\theta} = \sum_{i=1}^{I}\sum_{j=1}^{J_i} \rho_{ij}\underline{\theta}_{ij}$, $\theta_0 = \beta + \underline{\theta}$, and

$$\theta_m = \theta_0 + \sum_{l=1}^{m} \eta_l \text{ for } m = 1, \ldots, M.$$

Note that $\theta_M > 0$ because $\beta > 0$ by the heavy traffic assumption and that $\bar{\theta}_{ij} > 0$ for all ij. Also note that $0 < c_1 < c_2 < \ldots < c_M$ and define the piecewise linear, convex, and increasing function $c(\cdot)$ on $A = [\theta_0, \theta_M]$ as follows: $c(\theta_0) = 0$, and

$$c(x) = \sum_{i=1}^{m-1} c_i(\theta_i - \theta_{i-1}) + c_m(x - \theta_{m-1}),$$
$$\theta_{m-1} < x \leq \theta_m, \; m = 1, \ldots, M.$$

Figure 3. An Illustrative $c(\cdot)$ Function

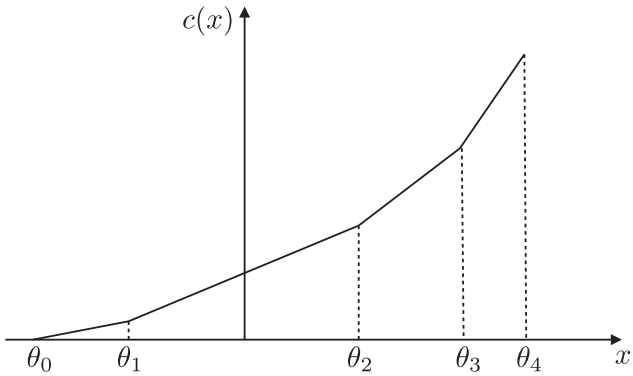


Figure 3 displays an illustrative $c(\cdot)$ function with $M = 4$.

Then consider the following minimization problem: for $s \geq 0$, choose $\theta(s) \in [\theta_0, \theta_M]$ so as to

$$\text{minimize } \overline{\lim}_{t\to\infty} \frac{1}{t}\, \mathbb{E}\left[\int_0^t \mu\, c(\theta(s))\, ds + \mu\, pU(t)\right]$$

subject to

$$\begin{aligned} &dZ(t) = \big[m(Z(t)) - \mu\theta(t)\big]dt + \sigma\, dB(t) - \mu dU(t), \quad t \geq 0, \\ &Z(t) \leq b, \quad t \geq 0, \qquad (P) \\ &\int_0^t \mathbb{1}_{\{Z(t)<b\}}\, dU(t) = 0, \quad t \geq 0, \\ &U(\cdot) \text{ is nondecreasing with } U(0) = 0. \end{aligned}$$

The following proposition establishes the equivalence of the formulations ($\tilde{P}$) and (P).

Proposition 2. *The problems ($\tilde{P}$) and (P) are equivalent in the following sense. For every feasible policy $\tilde{\theta}(\cdot)$ of ($\tilde{P}$), there exists a feasible policy $\theta(\cdot)$ for (P) that has lower (or the same) cost. In particular, given $\tilde{\theta}(\cdot)$ feasible for ($\tilde{P}$), one can define such a policy $\theta(\cdot)$ as follows:*

$$\theta(s) = \sum_{i=1}^{I} \sum_{j=1}^{J_i} \rho_{ij} \tilde{\theta}_{ij}(s) \text{ for } s \geq 0.$$

Similarly, every feasible policy $\theta(\cdot)$ of (P) yields a feasible policy $\tilde{\theta}(\cdot)$ for ($\tilde{P}$) that has the same cost. To be more specific, given the feasible policy $\theta(\cdot)$ for (P), one can define the corresponding policy $\tilde{\theta}(\cdot)$ for ($\tilde{P}$) as follows: for $s \geq 0$, if $\theta(s) = \theta_M$, then $\tilde{\theta}_{ij}(s) = \bar{\theta}_{ij}$ for all i, j. Otherwise, $\theta(s) \in [\theta_m, \theta_{m+1})$ for some $m = 0, 1, \ldots, M-1$. For each ij, letting $r_1(i,j) = rank(\pi^r_{ij})$ and $r_2(i,j) = rank(\pi^l_{ij})$, we have that

$$\tilde{\theta}_{ij}(s) = \begin{cases} \bar{\theta}_{ij} & \text{if } r_2(i,j) \leq m, \\ \frac{\theta(s)-\theta_m}{\rho_{ij}} & \text{if } r_2(i,j) = m+1, \\ 0 & \text{if } r_2(i,j) > m+1,\ r_1(i,j) \leq m, \\ \underline{\theta}_{ij} + \frac{\theta(s)-\theta_m}{\rho_{ij}} & \text{if } r_1(i,j) = m+1, \\ \underline{\theta}_{ij} & \text{if } r_2(i,j) > m+1. \end{cases} \qquad (15)$$

The next section solves the formulation (P) and characterizes the optimal policy as a nested threshold policy on Z. However, as discussed, the planner does not directly observe Z in practice. Nonetheless, the planner can observe the backlog of gleaning donations and, therefore, estimate the backlog of volunteer jobs by summing the expected number of gleaners (which is a function of the requested staffing level k) that will attend each of these backlogged donations, $\mathbb{E}[N(k)]$. In the heavy traffic regime, this approximation is not only asymptotically accurate, but it also leads to an implementable policy in practice. In Section 6, to assess the performance of the dynamic policy, we simulate the actual gleaning process in which Z is unobservable and use $\mathbb{E}[N(k)]$ to determine the staffing level and show that the dynamic policy still offers significant performance gains. Therefore, in what follows, we proceed as though the backlog process Z is observable to the planner.

5. Solution to the Limiting Brownian Control Problem

This section characterizes the optimal policy. To minimize technical complexity, we restrict attention to stationary, Markov control policies. That is, the (excess) staffing level requested at any time t is assumed to depend on past history only through the current backlog $Z(t)$. Thus, an admissible control policy is then a function $\theta(\cdot) : (-\infty, b] \to A$. To characterize the optimal policy analytically, consider the associated Bellman equation: find a twice-continuously differentiable, convex increasing function f and the constant $\gamma > 0$ that jointly satisfy the following (second order, nonlinear) differential equation:

$$\gamma = \min_{x\in A} \left\{\frac{1}{2}\sigma^2 f''(z) + (m(z) - \mu x) f'(z) + \mu c(x)\right\} \qquad (16)$$

subject to the boundary conditions

$$\lim_{z\to-\infty} f'(z) = 0 \qquad \text{and} \qquad f'(b) = p. \qquad (17)$$

We interpret γ as a guess at the optimal average cost, and the unknown function f is the so-called relative value function in average cost dynamic programming. The properties of the Bellman equation used in our analysis are proved from first principles (see Proposition 3 and Theorem 1).

To facilitate the analysis to follow, define the convex conjugate of the cost function $c(\cdot)$ as follows:

$$\phi(y) = \sup_{x\in A}\{yx - c(x)\}, \quad y \geq 0. \qquad (18)$$

It is straightforward to prove that there exists a smallest $x^* \in A$ that achieves the supremum, which is denoted by $\psi(y)$; that is,

$$\psi(y) = \inf \operatorname*{argmax}_{x\in A}\{yx - c(x)\}, \quad y \geq 0. \qquad (19)$$

See Online Appendix C.4 for further elaborations of the $\phi(\cdot)$ and $\psi(\cdot)$ functions.

Note that the Bellman Equations (16) and (17) do not involve the function f itself. So it is possible to reduce to a first order differential equation by setting $v(x) = f'(x)$ for $x \le b$. Then, using the definition of ϕ, we express the Bellman equation equivalently as follows: choose a nondecreasing, continuously differentiable function v and a constant $\gamma > 0$ that jointly satisfy the following:

$$\gamma = \frac{1}{2}\sigma^2 v'(z) + m(z)v(z) - \mu\phi(v(z)) \tag{20}$$

subject to the boundary conditions

$$\lim_{z\to-\infty} v(z) = 0 \qquad \text{and} \qquad v(b) = p. \tag{21}$$

The following proposition establishes the existence of a solution to the Bellman equation; see Online Appendix B for its proof. Online Appendix A provides a constructive proof and a rather explicit characterization of the function $v(\cdot)$. That characterization of $v(\cdot)$, in turn, facilitates the rather explicit description of the optimal policy in Section 5.2.

Proposition 3. *The Bellman and (21) have a solution (γ^*, v), where $\gamma^* > 0$ and v is increasing and continuously differentiable.*

Then, defining

$$f(x) = -\int_x^b v(y)dy, \qquad x \le b, \tag{22}$$

and noting that $f'(x) = v(x)$, the following corollary is immediate from Proposition 3 and it provides a solution to Equations (16) and (17).

Corollary 1. *The pair (f, γ^*) solves the Bellman (16) and (17).*

In each state $z \le b$, our candidate policy chooses

$$\theta^*(z) = \psi(v(z)). \tag{23}$$

Because $v(\cdot)$ is strictly increasing, it is invertible. Denoting that inverse by $v^{-1}(\cdot)$, let

$$\tau_i^* = v^{-1}(c_i),\ i = 1, 2, \ldots, M. \tag{24}$$

Then, it follows from Equation (19) (see also Equation (92) in Online Appendix C.4) that the candidate policy $\theta^*(\cdot)$ (see Equation (23)) is a nested threshold policy with thresholds $\tau_1^* < \tau_2^* < \ldots < \tau_M^* < b$. That is,

$$\theta^*(z) = \begin{cases} \theta_0, & \text{if } z < \tau_1^*, \\ \theta_m, & \text{if } \tau_m^* \le z < \tau_{m+1}^*,\ m = 1, \ldots, M-1, \\ \theta_M, & \text{if } \tau_M^* \le z \le b. \end{cases} \tag{25}$$

The following result establishes that the candidate policy is indeed optimal; see Online Appendix B for its proof.

Theorem 1. *The candidate policy $\theta^*(\cdot)$ given in Equation (25) is optimal and has an average cost of γ^*.*

Given the optimal thresholds $\tau_1^*, \ldots, \tau_M^*$ for the Brownian control problem (P), define

$$Q_j^* = n + \sqrt{n}\,\tau_j^* \text{ for } j = 1, \ldots, M.$$

Then recalling that $r_1(i,j) = \text{rank}(\pi_{ij}^r)$ and $r_2(i,j) = \text{rank}(\pi_{ij}^l)$ and combining Proposition 2, Theorem 1, and Equations (5) and (6) leads to the following proposed policy for the original system.

Proposed staffing policy for the multicrop gleaning operation: For $t \ge 0$ and class ij, we have the following three cases:

(1) $Q(t) < Q_1^*$, then set $k_{ij}(t) = \bar{k}_{ij}$.
(2) $Q(t) \ge Q_M^*$, then set $k_{ij}(t) = \underline{k}_{ij}$.
(3) $Q(t) \in [Q_m^*, Q_{m+1}^*)$ for $m = 1, \ldots, M-1$, then set

$$k_{ij}(t) = \begin{cases} \underline{k}_{ij} & \text{if } r_2(i,j) \le m, \\ \bar{k}_{ij} & \text{if } r_1(i,j) \ge m+1, \\ k_{ij}^* & \text{otherwise (i.e., } r_2(i,j) \ge m+1, r_1(i,j) \le m). \end{cases}$$

5.1. An Outline of the Solution to the Bellman Equation

Our analysis of the limiting Brownian control problem proceeds by studying the Bellman equation in two intervals separately. This is because the system behavior, namely the drift rate (without control) of the diffusion process that governs the evolution of the system, is rather different depending on whether the state is positive or negative. As observed also in Halfin and Whitt (1981), when the system state is positive, it evolves (locally) as a Brownian motion. In contrast, when the state is negative, it evolves (locally) as an Ornstein–Uhlenbeck process with a linear drift toward the origin. This, of course, causes the Bellman equation to behave differently in these two cases. Indeed, the analysis for $z \ge 0$ closely mimics that of Ata et al. (2005), whereas the analysis for $z < 0$ is novel. Moreover, whether one can "paste" the solutions for the two parts suitably (see Equation (28)) so as to solve the Bellman equation over the entire state space is unclear a priori. Therefore, a technical contribution of the paper is to construct a solution to the Bellman equation for $z \le 0$ and combine that with the solution for $z \ge 0$ in a smooth fashion. Along the way, we also come up with a rather explicit characterization of the optimal policy. To this end, let

$$\phi_* = -\inf_{0\le u\le p} \phi(u). \tag{26}$$

Also let

$$\underline{\gamma} = \mu\phi_* \quad \text{and} \quad c_* = \inf\{x \geq 0 : \phi(x) = -\phi_*\}. \tag{27}$$

Then, for $\gamma > \underline{\gamma}$, consider the following two initial value problems. The first one is on the negative real line and is denoted by IVP$_-(\gamma)$.

$$\begin{aligned} &\gamma = \frac{1}{2}\sigma^2 w'(z) - \mu z w(z) - \mu\phi(w(z)) \\ &\text{subject to } \lim_{z\to-\infty} w(z) = 0. \end{aligned} \tag{IVP$_-(\gamma)$}$$

The second one is on the positive interval $[0, b]$ and is denoted by IVP$_+(\gamma)$.

$$\begin{aligned} &\gamma = \frac{1}{2}\sigma^2 u'(z) - \mu\phi(u(z)) \\ &\text{subject to } u(b) = p. \end{aligned} \tag{IVP$_+(\gamma)$}$$

For each $\gamma > \underline{\gamma}$, we derive the solutions to both problems, thereby establishing the existence and uniqueness of a smooth solution for each by construction. These solutions are denoted by $w(\cdot, \gamma)$ and $u(\cdot, \gamma)$, respectively. Ultimately, we pin down the optimal long-run average cost γ^* by setting

$$u(0, \gamma^*) = w(0, \gamma^*), \tag{28}$$

and will set (the derivative of) the relative value function to be

$$v(z) = \begin{cases} w(z, \gamma^*), & z < 0, \\ u(z, \gamma^*), & z \in [0, b]. \end{cases} \tag{29}$$

Ensuring that Equation (28) has a unique solution and that (γ, v) given by Equations (28) and (29) solves the Bellman Equations (20) and (21) requires a careful analysis involving several steps, which also yields a rather explicit characterization of $v(\cdot)$ and the corresponding optimal policy, that is, the thresholds $\tau_1^*, \tau_2^*, \ldots, \tau_M^*$ as a by-product. Indeed, Section 5.2 provides a closed-form characterization of the thresholds $\tau_1^*, \tau_2^*, \ldots, \tau_M^*$ in terms of the optimal average cost γ^* and the problem primitives.

5.2. Further Characterization of the Thresholds and the Proposed Staffing Policy

The analysis outlined immediately in the last section yields an explicit characterization of the optimal thresholds $\tau_1^*, \ldots, \tau_M^*$. To facilitate this characterization, define the auxiliary function $H(v, \gamma)$ for $\gamma > \underline{\gamma}$ and $v \in [0, p]$ as follows:

$$H(v, \gamma) = b - \frac{\sigma^2}{2}\int_v^p \frac{dy}{\gamma + \mu\phi(y)}. \tag{30}$$

Also define g_i and G_i as the probability density function and cumulative distribution function, respectively, of a normal random variable with mean $-\theta_i$ and variance $\sigma^2/(2\mu)$ for $i = 0, 1, \ldots, M$. The characterization also involves the function $w(\cdot, \gamma^*)$. Proposition 6 shows that $w(\cdot, \gamma^*)$ is increasing on $(-\infty, 0]$; hence, it is invertible. Denote that inverse by $w^{-1}(\cdot, \gamma^*)$, whereas Propositions 5 and 9 provide explicit characterizations of it; see Online Appendix A.1 for the statements and the proofs of the propositions.

Proposition 4. *The following $M + 1$ cases characterize the thresholds of the optimal policy, $\tau_1^*, \ldots, \tau_M^*$:*

Case 0. $2\gamma^* \leq c_1\sigma^2\,(g_0(0)/G_0(0))$;

Case j. $(j = 1, \ldots, M-1)$ $2\gamma^* > c_1\sigma^2\,(g_0(0)/G_0(0))$, $H(c_{j+1}, \gamma^*) > 0$, *and* $H(c_j, \gamma^*) \leq 0$;

Case M. $2\gamma^* > c_1\sigma^2\,(g_0(0)/G_0(0))$ *and* $H(c_M, \gamma^*) \leq 0$.

In case 0, we have that $\tau_i = H(c_i, \gamma^*)$ for $i = 1, \ldots, M$. Moreover, in case j ($j = 1, \ldots, M$), we have the following two subcases[5]*:*

Case j(a). $\tau_i^* = H(c_i, \gamma^*)$ *for* $i = j+1, \ldots, M$, *and*

Case j(b). $\tau_i^* = w^{-1}(c_i, \gamma^*)$ *for* $i = 1, \ldots, j$.

In particular, in the subcase j(b), the thresholds $\tau_1^*, \ldots, \tau_j^*$ can be computed as follows:

$$\tau_1^* = \left(\frac{G_0}{g_0}\right)^{-1}\left(\frac{c_1\sigma^2}{2\gamma^*}\right) < 0. \tag{31}$$

Then given $\tau_{i-1}^* < 0$, the next threshold $\tau_i^* < 0$ for $i = 2, \ldots, j$ is computed by solving the following equation:

$$c_{i-1}\frac{g_{i-1}(\tau_{i-1}^*)}{g_{i-1}(\tau_i^*)} + \frac{2(\gamma^* - \mu c(\theta_{i-1}))}{\sigma^2}\frac{G_{i-1}(\tau_i^*) - G_{i-1}(\tau_{i-1}^*)}{g_{i-1}(\tau_i^*)} = c_i. \tag{32}$$

6. Numerical Study

To study the effectiveness of our proposed policy, we conduct a numerical study using parameters based on the process characteristics of the Boston Area Gleaners (2016), a gleaning organization that operates in the Boston metropolitan area. Currently, when deciding the staffing level for a gleaning trip, BAG primarily considers the type of crop and the characteristics of the farm (Caldwell et al. 2017). In the context of our model, this would correspond to a fixed staffing level for each class *ij* trip. Another staffing policy that is used by the Food Bank of the Southern Tier (FBST) (2015) in New York is to send out a broadcast email to the entire volunteer list for each gleaning trip and not set any limit on the number of volunteers that can attend each trip. We compare our dynamic staffing policy to the class-dependent static policy used by BAG and the unlimited policy used by FBST.

To assess the effectiveness of the dynamic policy, we construct a simulation model for the gleaning process that reflects the actual gleaning process flow (i.e., without making the simplifying assumptions used to derive the dynamic policy). Specifically there are four

simplifying assumptions made in the analytical model that we do not incorporate into the simulation. First, the payoff for each gleaning trip is calculated using the actual realizations for gleaner attendance, not the piecewise linear approximation shown in Figure 1. Second, instead of breaking donations into volunteer job flow units, the simulation uses donations as the flow unit so that volunteers attending the same trip are synchronized in their gleaning timing (as would be the case in practice). Third, if there are not enough available volunteers during the gleaning window to fill the requested staffing level, the gleaning trip proceeds with however many are available, and we lose a part of the payoff proportional to the number of missing volunteers. Fourth, because the planner cannot observe $N(k)$ of a backlogged donation until the gleaning trip occurs (note, however, that the planner can observe the backlog of gleaning donations), we use $\mathbb{E}[N(k)]$ instead of $N(k)$ to estimate the backlog for the purpose of deciding the staffing level. Thus, even though we used simplifying process assumptions to derive the dynamic policy, we assess the performance of this policy relative to the static and unlimited policies by applying them in a simulation model that reflects the reality of the gleaning process (without process simplifications).

In Online Appendix E, we compare our simulation output with the actual output of the 2014 BAG gleaning season and verify that the actual output measures are within the 95% confidence intervals of the simulation output measure. In the numerical analysis, we use our simulation model applied to the process parameters of the 2016 BAG gleaning season to compare the performance of our dynamic policy with static and unlimited policies. The 2016 season reflects the growth in the gleaning operation (increased number of donations and volunteers), and the dynamic policy can be helpful in managing system congestion.[6]

To compare the performance of the dynamic policy to the static and unlimited policies, we chose three crop types—corn, apple, and chard (indexed by $i = 1, 2$, and 3, respectively)—and use the distribution of these crop-type donations from 2016. Online Appendix D provides the values for specific gleaning parameters. We assumed that values for the three crop types are $\nu_1 = \nu_2 = \nu_3 = 1$ so that the payoffs represent the number of pounds gleaned. This is consistent with the way BAG reports its output and gives a straightforward interpretation of the results, but obviously ν_i can be set to translate pounds gleaned into another performance metric of interest, such as serving size or wholesale price. Additionally, by using pounds gleaned as the output metric, these very different crops with very different payoff functions illustrate some interesting properties of the dynamic policy.

We base the number of volunteers off of BAG's volunteer pool size in 2016: $n = 400$. We use an estimate from BAG of the rate at which volunteers become available as $\mu = 0.0253$ (i.e., each volunteer becomes available approximately every 40 days) (Caldwell et al. 2017). The productivity of a volunteer (r_i, the number of pounds gleaned per volunteer per trip) varies depending on the type of crop. These productivity numbers were estimated based on productivity numbers of actual gleaning trips of these crop types (see Online Appendix D).

Typically, when a farmer calls with a gleaning donation, the gleaning trip must be scheduled within a seven-day window; this implies that the backlog of jobs cannot exceed $b_n = 7n\mu \approx 70$ jobs. Although the data are not available to determine the probability that a gleaner will attend a trip, estimates from BAG indicate that this probability is fairly high (Caldwell et al. 2017); thus, we assume that $q = 0.8$.

We compare the dynamic policy to two policies used in practice: the unlimited policy and the static policy. The unlimited policy allows any number of volunteers to attend a gleaning trip. The gleaning organization simply sends out a trip announcement to the entire email list of volunteers, and whoever is available can attend the trip. Under the static policy, the gleaning organization posts a sign-up page with a specified number of volunteer slots for each gleaning trip. The number of slots depends on the donation class (crop type and donation size).

Clearly, the static-policy staffing level and the staffing levels in the dynamic policy (i.e., k^*_{ij}, $\underline{k}_{ij}$, and $\bar{k}_{ij}$ for each class ij) are parameters that can be optimized. However, we use straightforward heuristics to determine the values of these parameters in our simulation to reflect the process used by BAG. That is, we use a static staffing level that depends on the crop type and size of the donation with some safety capacity built in to compensate for volunteers who potentially may not show up for the gleaning trip. In particular, we use the static policy that sets the number of slots to $\tilde{k}_{ij}/q$ for all ij. For the staffing parameters of the dynamic policy, we take the static policy for each class as the nominal level and consider a range that is plus or minus 20% of the nominal, a reasonable range based on discussions with BAG (Caldwell et al. 2017). That is, we set the nominal staffing level in the dynamic policy to $k^*_{ij} = \tilde{k}_{ij}/q$ and the lower and upper bounds of staffing level to $\underline{k}_{ij} = k^*_{ij} - 0.2k^*_{ij}$ and $\bar{k}_{ij} = k^*_{ij} + 0.2k^*_{ij}$, respectively. We use the results from Proposition 4 to determine the optimal thresholds for the dynamic policy. The most interesting regime is when the system is congested, that is, in heavy traffic; therefore, we set the overall gleaning donation arrival rate to $\lambda^n = \sum_{ij} \lambda^n_{ij} = 0.93$ arrivals per day. The fraction of arrivals that are corn, apple, and

chard, respectively, are determined by the actual distribution of donations of these crop types during the 2016 gleaning season.

To compare the performance of the dynamic, static, and unlimited policies, we develop a discrete event simulation model of our setting using the Arena simulation software. We select the simulation run times and number of replications such that the estimate of a given performance metric of interest reaches steady state, and the half width of a 95% confidence interval on this estimate is at most 0.5% of the mean. Each simulation scenario is replicated 100 times using a replication run time of 110,000 days with the first 10,000 days used as warm-up. The steady state was confirmed by plotting the performance metric estimate versus the run time.

Figure 4 shows the structure of the dynamic policy for this example with three crop types and 10 classes. The requested staffing level for each class is shown as a function of the queue length $Q(t)$ with the thresholds $Q_1^*, \ldots, Q_M^*$, where $M = 20$, indicated on the x-axis. Notice that for each crop type i, as the donation size increases, the threshold queue length to switch from the nominal staffing level k_{ij}^* to the low staffing level $\underline{k}_{ij}$ increases, and the threshold queue length to switch from the nominal staffing level k_{ij}^* to the high staffing level $\bar{k}_{ij}$ decreases. We can see this by looking at, for example, the corn staffing levels in Figure 4. The solid arrow that indicates the queue lengths for which the nominal staffing level k_{ij}^* is optimal gets longer as the donation size increases. This is intuitive because $\pi_{i1}^l < \pi_{i2}^l < \pi_{i3}^l$ (see Online Appendix D), which means that, below the nominal staffing level, it is very costly for large-size trips to lose volunteers. Thus, the queue-length threshold for switching from k_{ij}^* to $\underline{k}_{ij}$ is higher for larger size trips. Similarly, $\pi_{i1}^r > \pi_{i2}^r > \pi_{i3}^r$, which means that above the nominal staffing level, the benefit of more volunteers decreases in donation size. Therefore, the queue length threshold for switching from k_{ij}^* to $\bar{k}_{ij}$ is lower for larger size trips.

Comparing the staffing levels across crop types can also be instructive. For example, for a given congestion level, chard staffing levels for almost all donation sizes are lower than apple staffing levels. Comparing the staffing levels for class 24 (large apple donation) and class 31 (small chard donation) in Figure 4, we see that at congestion level Q_4^*, the dynamic policy would switch to the low staffing level $\underline{k}_{31}$ for class 31 chard donations but still remain at the high staffing level $\bar{k}_{24}$ for class 24 apple donations. This is because the value (measured by number of pounds) of chard donations is much smaller than apple donations. Thus, the dynamic policy assigns higher staffing levels to the higher value apple donations. This, of course, could change under a different metric (e.g., serving size or wholesale price).

Figure 5 shows the performance of the dynamic policy compared with the static staffing policy. In particular, Figure 5 shows the percentage increase in net payoff (i.e., the total volume gleaned minus the total penalty for rejected donations) as a function of the penalty parameter p. Recall that the penalty for a rejected donation of class ij is $\tilde{p}_{ij} = pk_{ij}^*$, which is roughly proportional to the size of the donation. Our conversations with gleaning organizations indicate that a reasonable range for the penalty is approximately the volume of one trip. That is, if a farmer's gleaning donation is rejected, the loss of goodwill results in roughly the loss of one future gleaning donation. In fact, the penalty could be considerably higher, depending on the willingness of a farmer to engage with the gleaning organization. Making the extra effort to coordinate with the gleaning organization and allowing amateur pickers onto the fields create additional work and risk for the farmer. If, in

Figure 4. Structure of the Dynamic Policy

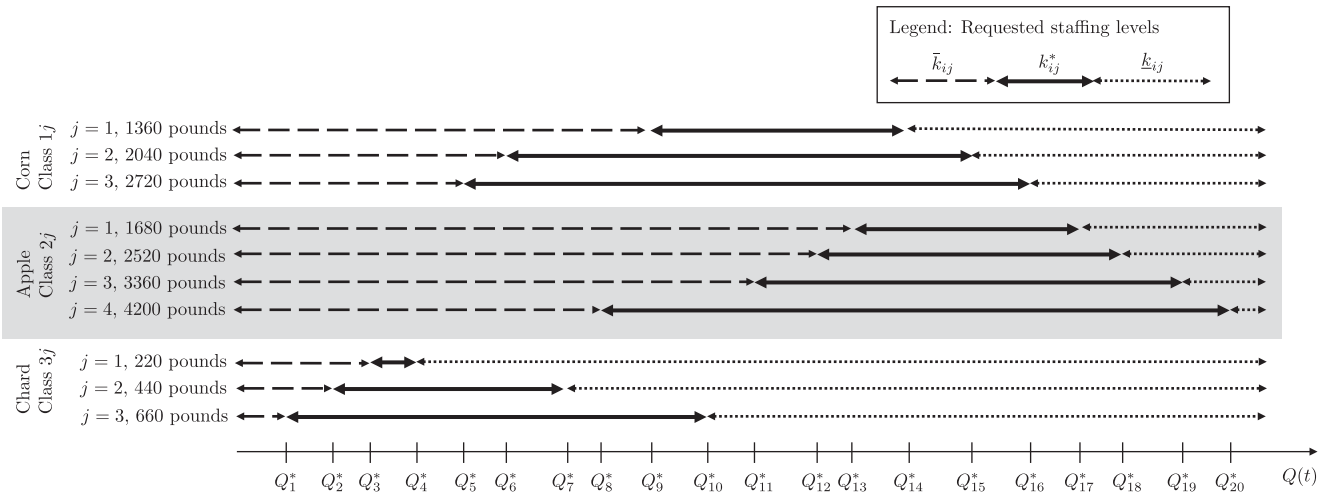
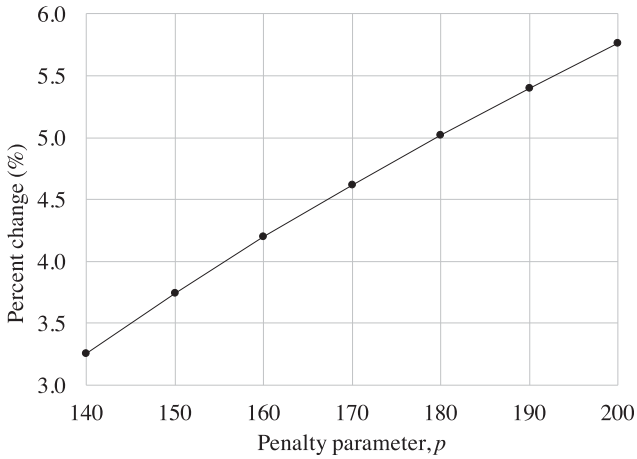


Figure 5. Percentage Increase in Net Payoff Achieved by the Dynamic Policy Compared with the Static Policy: $n = 400$, $q = 0.8$, $\mu = 0.0253$

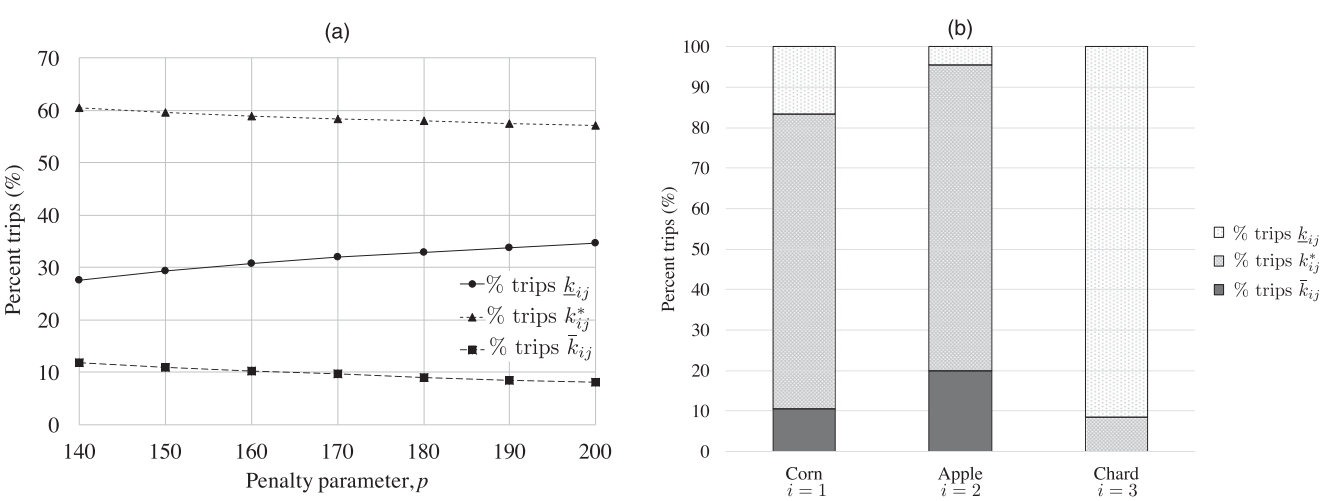


addition to those issues, the gleaning organization rejects well-intentioned donations, farmers may not find it worthwhile to participate in the gleaning network at all. Therefore, the gleaning organization needs to be responsive to farmers. To capture a range of penalty values, we show the performance of the dynamic policy relative to the static and unlimited policies for a range of p values.

Our simulation results show that the percentage increase in average net payoff of the dynamic policy (for staffing range $k^*_{ij} \pm 0.2k^*_{ij}$)[7] over the static policy ranges from slightly more than 3% to slightly less than 6% for penalty values ranging from $p = 140$ to $p = 200$, respectively (Figure 5). Moreover, the relative performance increases in p. This is intuitive because, as the penalty increases, the static cannot adjust for it, whereas the dynamic policy can. The percentage improvement of the dynamic policy over the unlimited policy is much higher—approximately 40%—because the unlimited policy is very inefficient in using volunteer capacity. Many trips are overstaffed, which then causes later trips to be understaffed, thus resulting in low overall volume gleaned. However, similar to the comparison with the static policy, the percentage improvement of the dynamic policy over the unlimited policy increases in the penalty parameter, p.

Figure 6 shows how the dynamic policy adjusts the staffing policy to improve performance. Figure 6(a) shows the percentage breakdown of trips for all classes combined as a function of the penalty. When the penalty is low, the percentage of rejected donations is relatively high. As the penalty increases, the dynamic policy shifts the staffing level allocation for each class ij from $\bar{k}_{ij}$ and k^*_{ij} to $\underline{k}_{ij}$ to save volunteers for later trips. This reduces the average number of donations in the queue, thereby reducing the probability of having to reject future gleaning donations, thus avoiding the penalty. Figure 6(b) shows the trip breakdown by crop for $p = 150$. Recall that because $\nu_1 = \nu_2 = \nu_3 = 1$, the heavier crops (i.e., corn and apples) are higher value crops than the lighter crop, chard. Thus, we see that the dynamic policy mostly allocates k^*_{ij} and $\bar{k}_{ij}$ staffing levels to the heavier crops, whereas it allocates mostly the $\underline{k}_{ij}$ staffing level for the lighter chard crop. In fact, during the simulation, there was not an instance of the high staffing level $\bar{k}_{ij}$ being assigned for chard. This is because whenever a chard donation arrived (i.e., a class 31, 32, or 33 arrival), the queue length was always greater than the threshold for staffing level $\bar{k}_{3j}$, $j = 1, 2, 3$. Figure 4 shows that the thresholds corresponding to $\bar{k}_{31}$, $\bar{k}_{32}$, and $\bar{k}_{33}$ are the lowest thresholds, Q^*_1, Q^*_2, and Q^*_3, respectively. Thus, the staffing levels $\bar{k}_{31}$, $\bar{k}_{32}$, and $\bar{k}_{33}$ would not be assigned unless the system congestion was quite low—a

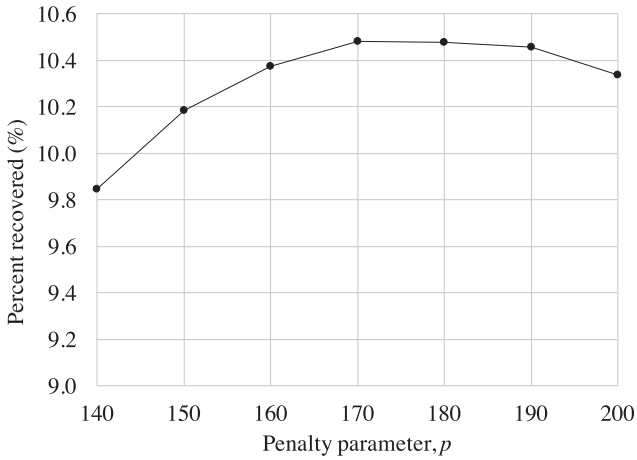
Figure 6. Percentage Breakdown of Trips at Each Staffing Level for the Dynamic Policy: $n = 400$, $q = 0.8$, $\mu = 0.0253$



All crop types combined

By crop type for $p = 150$

Figure 7. Percentage of Static Policy Volume Loss Recovered Under the Dynamic Policy: $n = 400$, $q = 0.8$, $\mu = 0.0253$



condition that was not satisfied whenever a chard donation arrived.

Note that the net payoff includes the penalty cost for rejecting donations (to account for loss of farmer goodwill). Another important performance metric is how much of the volume lost under the static or unlimited policy can be recovered by using the dynamic policy. Figure 7 shows the dynamic policy can recover approximately 10% of the volume lost under the static policy. The percentage recovered starts to decrease for higher penalties because the policy emphasizes reducing rejections; therefore, more trips are assigned $\underline{k}_{ij}$ staffing levels, thereby reducing the volume gleaned per trip. If we break down the recovered volume by crop, we see that the recovered corn and apple crops come at the expense of chard. For example, for $p = 150$, the dynamic policy recovers 6.4% and 25.3% of the volume of corn and apples, respectively, that are lost under the static policy. However, the volume of chard gleaned under the dynamic policy is only 88.8% of the volume gleaned under the static policy. Thus, even though the total volume gleaned increases under the dynamic policy, at the crop level, the volume changes could be positive or negative. Additionally, compared with the unlimited policy, the percentage of the volume loss recovered by the dynamic policy is considerably higher—in the range of 74%–75%—because the volume loss under the unlimited policy is so much higher.

Our numerical study parameters represent an organization approximately the size of the Boston Area Gleaners. To put these recovery numbers in perspective, running a static policy similar to BAG's policy would imply an annual volume loss of 46,328 pounds. In this case, the dynamic policy would recover 4,633 pounds or an equivalent of 18,532 four-ounce servings. To achieve this increase in volume, no capital or major process changes would be required—only some small changes to the staffing level requests.

7. Conclusion

Organizing gleaning operations is a worthwhile endeavor that benefits society by alleviating food insecurity and reducing food waste. The gleaning operation presents an interesting operational challenge because the demand for gleaning services (the arrival of donations from farms) and the labor supply (volunteer gleaner capacity) are both stochastic. This uncertainty is particularly challenging when the demand rate and gleaner capacity are close. The gleaning organization needs to trade off safety capacity on the current trip with gleaner capacity for future trips.

This paper characterizes the gleaning process under the heavy traffic assumption and derives a (almost explicit) staffing policy that depends on the congestion in the system and allows for multiple donation classes (for different crop types and donation sizes). This policy is a nested threshold policy that specifies the staffing level for each donation class. In a numerical study with parameters for volunteer and donation characteristics calibrated using data from a gleaning organization in the Boston area, we found that the dynamic staffing policy can recover approximately 10% of the volume lost under the static policy and approximately 75% of the volume lost under the unlimited policy. The dynamic policy also increases the net payoff, and this improvement increases in the rejection penalty. Thus, our analysis can help gleaning organizations increase the volume gleaned by changing their staffing policy based on the congestion in the process. Moreover, these improvements do not require major investments or process changes—only small adjustments to the staffing request are required. We also provide a methodological contribution by explicitly solving the associated drift-rate control problem.

Our focus in this study was on volunteer staffing under congestion; however, there are several avenues of future research that can enrich our understanding of gleaning operations. First, we assumed that there would be demand for whatever is gleaned. However, the outlets for some crops, even if they are free, are limited. Therefore, the model could be extended to consider the demand side of the gleaning operation. Second, another resource that can constrain the output of the gleaning operation is the paid staff who coordinate and lead the gleaning trips. The number of trips that can be scheduled also depends on the capacity of the paid staff. This constraint can be added to capture another realistic dimension of the gleaning operation.

Acknowledgments

The authors thank Laurie Caldwell, Dylan Frazier, and Matt Crawford at the Boston Area Gleaners for providing data to

calibrate the parameters of the model and for sharing valuable information about their gleaning operations.

Endnotes

[1] Our definition allows time-dependent infinitesimal drift and variance. So it is slightly more general. However, we ultimately focus attention on stationary Markov control policies; see Section 4. Hence, our definition coincides with that of Karlin and Taylor (1981).

[2] Moreover, by virtue of the state-space collapse results, imposing an overall delay constraint as done in Equation (7) also imposes the same delay constraint for every class ij separately; see Online Appendix C for a justification of this.

[3] The weak convergence result assumed in Equation (9) can be proved by combining Stone's lemma (see, for example, theorem 3.2 of Iglehart 1965) and the continuous mapping theorem for weak convergence (see Billingsley 1999), provided a certain regulator map (similar to the one given in Equations (10), (12), and (13)) is continuous. Reed and Ward (2004) and Reed et al. (2013) establish various useful properties of a closely related regulator map, including its Lipschitz continuity. The results of Reed and Ward (2004) and Reed et al. (2013) do not apply to our setting because we allow discontinuous drift terms in Equation (10). However, one may be able to extend their results to our setting, which would pave the way for proving a rigorous heavy traffic limit theorem. The study of these intricate technical issues is beyond the scope of this paper.

[4] Note that π^r_{ij} corresponds to the left slope, whereas π^l_{ij} corresponds to the right slope as shown in Figure 2.

[5] For the case $j = M$, we only have the subcase (b).

[6] We use the BAG data from 2014 to compare the simulation and actual outputs because the data from this year include donation arrivals and volunteer attendance on gleaning trips. Thus, we were able to compare the simulation and actual outputs for pounds gleaned and volunteer attendance. Moreover, we could use these data to estimate the volunteer productivity for various crops. In 2016, we only have the donation arrival data, but these more recent data better reflect the distribution of the crops and the growth in the operation. Therefore, in the simulation study, we use the 2016 donation data to compare the performance of the dynamic policy to the static and unlimited policies and assume that volunteer service rates and productivity remain the same as in 2014.

[7] Online Appendix F presents a sensitivity analysis and discussion of how the staffing level range affects the policy performance.

References

Altay N, Green WG (2006) OR/MS research in disaster operations management. *Eur. J. Oper. Res.* 175(1):475–493.

Ata B (2005) Dynamic power control in a wireless static channel subject to a quality of service contract. *Oper. Res.* 53(5): 842–851.

Ata B (2006) Dynamic control of a multiclass queue with thin arrival streams. *Oper. Res.* 54(5):876–892.

Ata B, Kumar S (2005) Heavy traffic analysis of open processing networks with complete resource pooling: Asymptotic optimality of discrete review policies. *Ann. Appl. Probab.* 15(1A):331–391.

Ata B, Shneorson S (2006) Dynamic control of an M/M/1 service system with adjustable arrival and service rates. *Management Sci.* 52(11):1778–1791.

Ata B, Tongarlak M (2013) On scheduling a multiclass queue with abandonments under general delay costs. *Queueing Systems* 74(1):65–104.

Ata B, Zachariadis K (2007) Dynamic power control in a fading downlink channel subject to an energy constraint. *Queueing Systems* 55(1):41–69.

Ata B, Harrison JM, Shepp LA (2005) Drift rate control of a Brownian processing system. *Ann. Appl. Probab.* 15(2):1145–1160.

Banerjee S, Riquelme C, Johari R (2015) Pricing in ride-share platforms: A queueing-theoretic approach. Working paper, Cornell University, Ithaca, NY. Available at SSRN: https://ssrn.com/abstract=2568258 or http://dx.doi.org/10.2139/ssrn.2568258.

Bell SL, Williams R (2001) Dynamic scheduling of a system with two para servers in heavy traffic with resource pooling: Asymptotic optimality. *Ann. Appl. Probab.* 11(3):608–649.

Billingsley P (1999) *Convergence of Probability Measures*, 2nd ed. (Wiley, New York).

Bollapragada S, Morton TE (1999) Myopic heuristics for the random yield problem. *Oper. Res.* 47(5):713–722.

Boston Area Gleaners (2016) Boston Area Gleaners. Accessed September 25, 2016, http://www.bostonareagleaners.org.

Browne S, Whitt W (1995) Piecewise-linear diffusion processes. Dshalalow JH, ed. *Advances in Queueing Theory, Methods, and Open Problems*, *Probability and Statistics Series* (CRC Press, Boca Raton, FL), 463–480.

Caldwell L, Frazier D, Crawford M (2017) Personal communication, April 17.

California Association of Food Banks (2011) Utilizing new methods of crop harvesting to introduce nutrient-dense specialty crops to low income consumers. Accessed May, 2015, http://www.cafoodbanks.org/sites/default/files/concurrent-harvesting-report.pdf.

Coleman-Jensen A, Rabbitt MP, Gregory C, Singh A (2015) Household food security in the United States in 2014. ERS Report 215, U.S. Department of Agriculture, Washington, DC.

Crabill T (1972) Optimal control of a service facility with variable exponential service times and constant arrival rate. *Management Sci.* 18(9):560–566.

Crabill T (1974) Optimal control of a maintenance system with variable service rates. *Oper. Res.* 22(4):736–745.

Davis B, Tarasuk V (1994) Hunger in Canada. *Agriculture Human Values* 11(4):50–57.

Davis LB, Jiang SX, Morgan SD, Harris C (2013) Forecasting donated goods for a local food bank. *Proc. Southeast Decision Sci. Inst. 2013 Annual Meeting* (Southeast Decision Sciences Institute, Charleston, SC), 95–103.

Davis LB, Sengul I, Ivy JS, Brock LG III, Miles L (2014) Scheduling food bank collections and deliveries to ensure food safety and improve access. *Socio-Econom. Planning Sci.* 48(3):175–188.

Dong J, Ibrahim R (2017) Flexible workers or full-time employees? On staffing systems with a blended workforce. Working paper, Columbia University, New York. Available at SSRN: https://ssrn.com/abstract=2971841.

Falasca M, Zobel C (2012) An optimization model for volunteer assignments in humanitarian organizations. *Socio-Econom. Planning Sci.* 46(4):250–260.

Food Bank of the Southern Tier (2015) Food Bank of the Southern Tier. Accessed December 6, 2015, http://www.foodbankst.org.

Galindo G, Batta R (2013) Review of recent developments in OR/MS research in disaster operations management. *Eur. J. Oper. Res.* 230(2):201–211.

George JM, Harrison JM (2001) Dynamic control of a queue with adjustable service rate. *Oper. Res.* 49(5):720–731.

Gerchak Y, Vickson RG, Parlar M (1988) Periodic review production models with variable yield and uncertain demand. *IIE Trans.* 20(2):144–150.

Ghamami S, Ward AR (2013) Dynamic scheduling of a two-server parallel server system with complete resource pooling and reneging in heavy traffic: Asymptotic optimality of a two-threshold policy. *Math. Oper. Res.* 38(4):761–824.

Ghosh A, Weerasinghe A (2010) Optimal buffer size and dynamic rate control for a queueing network with reneging in heavy traffic. *Stochastic Processes Their Appl.* 120(11):2103–2141.

Giffler B (1960) Determining an optimal reject allowance. *Naval Res. Logist. Quart.* 7(2):201–206.

Gomez M, Barrett C, Raney T, Pinstrup-Andersen P, Meerman J, Croppenstedt A, Carisma B, Thompson B (2013) Post-green revolution food systems and the triple burden of malnutrition. *Food Policy* 42(October):129–138.

Gordon L, Erkut E (2004) Improving volunteer scheduling for the Edmonton Folk Festival. *Interfaces* 34(5):367–376.

Gunders D (2012) Wasted: How America is losing up to 40 percent of its food from farm to fork to landfill (National Resources Defense Council). NRDC Issue Paper 12-06-B. Accessed May 18, 2015, http://www.nrdc.org/food/files/wasted-food-IP.pdf.

Gunes C, van Hoeve W-J, Tayur S (2010) Vehicle routing for food rescue programs: A comparison of different approaches. Lodi A, Milano M, Toth P, eds. *Integration of AI and OR Techniques in Constraint Programming for Combinatorial Optimization Problems*. Lecture Notes in Computer Science, vol. 6140 (Springer, Berlin Heidelberg), 176–180.

Halfin S, Whitt W (1981) Heavy-traffic limits for queues with many exponential servers. *Oper. Res.* 29(3):567–588.

Harrison J (1988) Brownian models of queueing networks with heterogenous customer populations. Fleming W, Lions PL, eds. *Stochastic Differential Systems, Stochastic Control Theory and Applications* (Springer-Verlag, New York), 147–186.

Harrison J (1998) Heavy traffic analysis of a system with parallel servers: Asymptotic optimality of discrete review policies. *Ann. Appl. Probab.* 8(3):822–848.

Harrison J, Wein L (1989) Scheduling network of queues: Heavy traffic analysis of a simple open network. *Queueing Systems* 5(4):265–280.

Harrison J, Wein L (1990) Scheduling networks of queues: Heavy traffic analysis of a two-station closed network. *Oper. Res.* 38(6):1052–1064.

Henig M, Gerchak Y (1990) The structure of periodic review policies in the presence of random yield. *Oper. Res.* 38(4):634–643.

Hoisington A, Butkus SN, Garrett S, Beerman K (2001) Field gleaning as a tool for addressing food security at the local level: Case study. *J. Nutrition Ed.* 33(1):43–48.

Hopp WJ, Iravani SMR, Yuen GY (2007) Operations systems with discretionary task completion. *Management Sci.* 53(1):61–77.

Huang W, Kulkarni V, Swaminathan JM (2008) Managing the inventory of an item with a replacement warranty. *Management Sci.* 54(8):1441–1452.

Huh WT, Nagarajan M (2010) Linear inflation rules for the random yield problem: Analysis and computations. *Oper. Res.* 58(1):244–251.

Ibrahim R (2018) Managing queueing systems where capacity is random and customers are impatient. *Production Oper. Management* 27(2):234–250.

Iglehart D (1965) Limit diffusion approximations for the many-server queue and the repairman problem. *J. Appl. Probab.* 2(2):411–429.

Inderfurth K, Kiesmuller GP (2015) Exact and heuristic linear-inflation policies for an inventory model with random yield and arbitrary lead times. *Eur. J. Oper. Res.* 245(1):109–120.

Inderfurth K, Transchel S (2007) Note on "myopic heuristics for the random yield problem". *Oper. Res.* 55(6):1183–1186.

Inderfurth K, Vogelgesang S (2013) Concepts for safety stock determination under stochastic demand and different types of random production yield. *Eur. J. Oper. Res.* 224(2):293–301.

Irwin JD, Ng VK, Rush TJ, Nguyen C, He M (2007) Can food banks sustain nutrient requirements? A case study in southwestern Ontario. *Canadian J. Public Health* 98(1):17–20.

Johnson MP, Smilowitz K (2007) Community-based operations research. Klastorin T, Gray P, eds. *OR Tools and Applications: Glimpses of Future Technologies*, INFORMS TutORials in Operations Research (INFORMS, Catonsville, MD), 102–123.

Karlin S (1958) One stage models with uncertainty. Arrow KJ, Karlin S, Scarf H, eds. *Studies in the Mathematical Theory of Inventory and Production* (Stanford University Press, Palo Alto, CA), 109–134.

Karlin S, Taylor H (1981) *A Second Course in Stochastic Processes* (Academic Press, New York).

Kim J, Ward AR (2013) Dynamic scheduling of a GI/GI/1 +GI queue with multiple customer classes. *Queueing Systems* 75(2–4): 339–384.

Kovacs G, Spens KM (2011) Trends and developments in humanitarian logistics – a gap analysis. *Internat. J. Physical Distribution Logist. Management* 41(1):32–45.

Lee D, Sönmez E, Gómez MI, Fan X (2017) Combining two wrongs to make two rights: Mitigating food insecurity and food waste through gleaning operations. *Food Policy* 68(April):40–52.

Lien RW, Iravani SM, Smilowitz KR (2014) Sequential resource allocation for nonprofit operations. *Oper. Res.* 62(2):301–317.

Mazzola JB, McCoy WF, Wagner HM (1987) Algorithms and heuristics for variable-yield lot sizing. *Naval Res. Logist.* 34(1):67–86.

Mohan S, Gopalakrishnan M, Mizzi PJ (2013) Improving the efficiency of a non-profit supply chain for the food insecure. *Internat. J. Production Econom.* 143(2):248–255.

Ormeci-Matoglu M, Vande Vate J (2011) Drift control with changeover costs. *Oper. Res.* 59(2):427–439.

Plambeck E, Kumar S, Harrison S (2001) A multiclass queue in heavy traffic with throughput time constraints: Asymptotically optimal dynamic controls. *Queueing Systems* 39(1):23–54.

Reed JE, Ward AR (2004) A diffusion approximation for a generalized Jackson network with reneging. *Proc. 42nd Allerton Conf. Comm. Control Comput.* (Curran Associates, Red Hook, NY).

Reed J, Ward A, Zhan D (2013) On the generalized drift Skorokhod problem in one dimension. *J. Appl. Probab.* 50(1):16–28.

Rubino M, Ata B (2009) Dynamic control of a make-to-order, parallel-server system with cancellations. *Oper. Res.* 57(1):94–108.

Sampson SE (2006) Optimization of volunteer labor assignments. *J. Oper. Management* 24(4):363–377.

Schuelke R, Hoffmann M, Gómez M (2011) Cause marketing opportunities: GleanNY, donating food from farms to NY's hungry. Smart Marketing. Accessed May 18, 2015, http://publications.dyson.cornell.edu/docs/smartMarketing/pdfs/SmrtMkg%20July2012.pdf.

Shih W (1980) Optimal inventory policies when stockouts result from defective products. *Internat. J. Production Res.* 18(6):677–685.

Solak S, Scherrer C, Ghoniem A (2014) The stop-and-drop problem in nonprofit food distribution networks. *Ann. Oper. Res.* 221(1):407–426.

Sönmez E, Lee D, Gómez MI, Fan X (2016) Improving food bank gleaning operations: An application in New York state. *Amer. J. Agricultural Econom.* 98(2):549–563.

Starr M, Van Wassenhove L (2014) Humanitarian operations and crisis management. *Production Oper. Management* 23(6):925–937.

Stidham S (1988) Scheduling, routing, and flow control in stochastic networks. Fleming W, Lions PL, eds. *Stochastic Differential Systems, Stochastic Control Theory and Applications* (Springer-Verlag, New York), 529–561.

Stidham S (2002) Analysis, design, and control of queueing systems. *Oper. Res.* 50(1):197–216.

Stidham S, Weber RR (1989) Monotonic and insensitive optimal policies for control of queues with undiscounted costs. *Oper. Res.* 37(4):611–625.

Tarasuk V, Beaton G (1999) Household food insecurity and hunger among families using food banks. *Canadian J. Public Health* 90(2): 109–113.

Taylor T (2018) On-demand service platforms. *Manufacturing Service Oper. Management* 20(4):704–720.

Teron AC, Tarasuk VS (1999) Charitable food assistance: What are food bank users receiving? *Canadian J. Public Health* 90(6):382–384.

USDA (2011) Crop production annual summaries (principal crops and vegetables). U.S. Department of Agriculture National Agricultural Statistics Service. Accessed June 12, 2016, http://usda.mannlib.cornell.edu/MannUsda/viewDocumentInfo.do?documentID=1047.

USDA (2015) Definitions of food security. U.S. Department of Agriculture. Accessed December 19, 2015, http://www.ers.usda.gov/topics/food-nutrition-assistance/food-security-in-the-us/definitions-of-food-security.aspx.

Van Wassenhove LN (2006) Humanitarian aid logistics: Supply chain management in high gear, Blackett memorial lecture. *J. Oper. Res. Soc.* 57(5):475–489.

Vitiello D, Grisso JA, Whiteside KL, Fischman R (2015) From commodity surplus to food justice: Food banks and local agriculture in the United States. *Agriculture Human Values* 32(3):419–430.

Weber RR, Stidham S (1987) Optimal control of service rates in network of queues. *Adv. Appl. Probab.* 19(1):202–218.

Wein L (1991) Brownian networks with discretionary routing. *Oper. Res.* 39(2):322–340.

Weinfield NS, Mills G, Borger C, Maeve G, Macaluso T, Montaquila J, Zedlewski S (2014) Hunger in America 2014. Feeding America. Accessed January 6, 2016, http://www.feedingamerica.org/hunger-in-america/our-research/hunger-in-america/.

Yano CA, Lee HL (1995) Lot sizing with random yields: A review. *Oper. Res.* 43(2):311–334.

Yildiz H, Johnson MP, Roehrig S (2013) Planning for meals-on-wheels: Algorithms and application. *J. Oper. Res. Soc.* 64(10): 1540–1550.

Zipkin PH (2000) *Foundations of Inventory Management* (McGraw-Hill, New York).

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